

**INTEGRITY MONITORING THROUGH RECEIVER
AUTONOMOUS INTEGRITY MONITORING (RAIM) FOR SINGLE
FREQUENCY GNSS APPLICATIONS**



by

Sunnyaha Saeed

Supervisor

Dr. Salma Zainab Farooq

Department of Space Sciences
Institute of Space Technology,
Islamabad, Pakistan
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Integrity Monitoring through Receiver Autonomous Integrity Monitoring (RAIM) for single frequency GNSS applications

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by
Sunnyaha Saeed

Supervisor
Dr. Salma Zainab Farooq

Department of Space Sciences,
Institute of Space Technology, Islamabad, Pakistan
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Institute of Space Technology
Department of Space Science



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Monitoring (RAIM) for single frequency GNSS applications**

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Sunnyaha Saeed

APPROVAL BY BOARD OF EXAMINERS

.....

Dr. Salma Zainab Farooq

Dr. Najam Abbas
Naqvi
(Internal Examiner)
Dept. of Space Science,
Institute of Space
Technology,
Islamabad, Pakistan

Dr.-----
(External Examiner)
Dept. -----

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ABSTRACT

Single frequency receivers for Global Navigation Satellite Systems (GNSS) are low in cost and easily accessible, deeming them cost effective for commercial autonomous applications such as fleet management and monitoring, drone photography, drone delivery, GNSS based tracking etc. However, the solution provided by these receivers must conform to desired reliability standards. Reliability check is particularly important in signal-degraded environment, where position fix might be unacceptably inaccurate. Under such conditions faulty measurement need to be identified and excluded to ensure integrity of the system and get a reliable solution. In this Research, a receiver autonomous integrity monitoring (RAIM) scheme is devised that uses two testing techniques: Least square residual and correlation analysis methods, faulty measurements are identified and excluded if both point to the same satellite. The exclusion of faulty satellite measurements allows reliable position fix using single frequency GNSS receiver. As a reliable indicator of the performance of positioning systems, Horizontal Protection Level (HPL) is also monitored through the developed algorithm. Both static and kinematic data sets are used to test the Scheme. Finally, selected findings and conclusions from these investigations are discussed.

Keywords: GNSS, GPS, Single Frequency Receiver, RAIM, Fault Detection, Fault Exclusion, FDE, Horizontal Protection Level

TABLE OF CONTENTS

AUTHOR’S DECLARATION	iv
CERTIFICATE	vi
Copyright © 2022	vii
ACKNOWLEDGMENT	viii
ABSTRACT	ix
TABLE OF CONTENTS	x
LIST OF TABLES	xiv
LIST OF FIGURES	xv
LIST OF ABBREVIATIONS	xvii
1. Introduction to GNSS	1
1.1 GNSS Segments	2
1.1.1 Space Segment	3
1.1.2 Control Segment	5
1.1.3 User Segment	6
1.2 GNSS Positioning Principle	7
1.3 GNSS Signals	8
1.3.1 GNSS Message Structure	9
1.3.2 Ranging Codes (C/A)	10

1.4	GNSS Error Sources.....	12
1.4.1	Satellite Clock Error	13
1.4.2	Ephemeris Error	13
1.4.3	Ionosphere and Troposphere Effects	14
1.4.4	Sagnac Effect	15
1.4.5	Multipath.....	16
1.4.6	Receiver Clock Errors.....	17
1.5	Doppler Effect	18
1.6	Dilution of Precision	18
1.7	GNSS Performance Parameters	19
i)	Availability.....	19
ii)	Continuity	19
iii)	Integrity	20
iv)	Accuracy	20
1.8	Problem Statement	20
1.9	Motivation	21
2.	Literature Review and Background	22
2.1	Navigation Solution Determination	23
2.1.1	Least Squares Navigation Solution.....	24
2.1.2	Weighted Least Square Navigation Solution.....	25

2.2	Receiver Autonomous Integrity Monitoring (RAIM).....	26
2.2.1	Filtering Scheme	26
2.2.2	Snapshot Scheme	26
2.3	Fault Detection and Exclusion	29
2.4	RAIM Algorithm.....	30
2.4.1	Methodology	30
2.4.2	Fault Detection Method	32
2.4.3	Protection level	33
2.4.4	Fault identification testing	34
3.	Data Collection and Experimental Setup.....	38
3.1	Experimental Setup	38
3.2	Data Collection.....	39
3.3	Algorithm Implementation.....	43
3.3.1	RINEX Files.....	43
4.	Results and Discussion	47
4.1	Experiment and Results of Dataset-1	47
4.2	Experiment and Results of Dataset-2	51
4.3	Experiment and Results of Dataset-3	54
5.	Summary and Future Research Direction	59
5.1	Summary	59

5.2	Future Research Direction.....	60
6.	References.....	61

LIST OF TABLES

Table 1.1 Specification of GNSS constellations.....	5
Table 1.2 Code Sequence.....	11
Table 3.1 Datasets Details.....	41
Table 4.1 Residual and Fault identification values for error introduced at epoch 400 for dataset-1	50
Table 4.2 Identification Rate (error >40m).....	Error! Bookmark not defined.
Table 4.3 Identification Rate (error <40m).....	Error! Bookmark not defined.

LIST OF FIGURES

Figure 1.1 GNSS Segments	2
Figure 1.2 Positioning Principle: At least Four Satellites are required to estimate four unknown [15]	7
Figure 1.3 Message Structure [16].....	10
Figure 1.4 Propagation Delays [17]	15
Figure 1.5 Sagnac Effect [17]	16
Figure 1.6 Multipath Error [17]	17
Figure 1.7 Satellite Geometry and DOP [19].....	19
Figure 2.1 Integrity Monitoring Taxonomy	22
Figure 2.2 Fault Detection	34
Figure 2.3 Fault Identification	37
Figure 3.1. GNSS Receiver.....	40
Figure 3.2 Data Collection Sites	41
Figure 3.3 Satellite Visibility Sky Plot(Dataset-1)	42
Figure 3.4 Satellite Visibility Sky Plot(Dataset-2)	42
Figure 3.5 Satellite Visibility Sky Plot(Dataset-3)	43
Figure 3.6 RINEX nav file.....	44
Figure 3.7 RINEX obs file	45
Figure 4.1 Global Test (Dataset -1)	48
Figure 4.2 Protection level vs Test Statistics	48
Figure 4.3 Fault Identification test: LSR- Test, Correlation Coefficient, and Correlation Distance (Dataset-1).....	49

Figure 4.4 Position with and without error	51
Figure 4.5 Global Test (Dataset-2)	52
Figure 4.6 Protection Level vs Test Statistics (Dataset-2).....	52
Figure 4.7 Fault Identification test: LSR- Test, Correlation Coefficient and Correlation Distance (Dataset-2).....	53
Figure 4.8 Position with and without Error	54
Figure 4.9 Global Test (Dataset-3)	55
Figure 4.10 PL vs SSE (Dataset-3).....	55
Figure 4.11 Fault Identification Tests (Dataset-3).....	56
Figure 4.12 Position with error and after applying RAIM.....	56

LIST OF ABBREVIATIONS

GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
GLONASS	Global'naya Navigatsionnaya Sputnikova Sistema
US	United States
QZSS	Quasi-Zenith Satellite System
IRNSS	Indian Regional Navigation System
MEO	Medium-Earth Orbit
LEO	Lower-Earth Orbit
HEO	Highly Elliptical Orbit
C/A	Coarse/Acquisition Code
P(Y)	Precision Code
BDS	BeiDou navigation satellite system
GEO	Geosynchronous Earth Orbit
IGSO	Inclined-Geosynchronous Earth Orbit
CDMA	Code Division Multiple Access
TLM	Telemetry
SV	Satellite Vehicle
LFSRs	Linear Feedback Shift Registers
XOR	exclusive-or
ZTD	Zenith Total Delay
DOP	Dilution of Precision
GDOP	Geometric Dilution of Precision
RAIM	Receiver Autonomous Integrity Monitoring
FDE	Fault Detection and Exclusion
GBAS	Ground Based Augmentation System
SBAS	Satellite Based Augmentation System
GIC	Ground Integrity Channel
ICAO	International Civil Aviation Organization
P_{FA}	Probabilities of false alarms
P_{MD}	Probabilities of missed detection
PL	Protection Level
VPL	Vertical Protection Level
HPL	Horizontal Protection Level
LSR	Least Square Residual
F-distribution	Fisher-distribution
RINEX	Receiver Independent Navigation Exchange format

ASCII	American Standard Code for Information Interchange
SSE	Sum of Squared Residuals

1. Introduction to GNSS

Navigation is the science of getting an object or person from one place to another. Over the centuries, various methods have been developed for determining directions on Earth's surface such as celestial navigation, use of landmarks, cross staff, astrolabe, etc. [1], [2]. With the passage of time new methods were invented for navigation. Since twentieth century, satellite-based positioning, navigation, and timing (PNT) technology has gained much attention due to its flexibility and ease of use [3], [4]. In the early 1970s, satellite-based navigation was initiated, and three systems were developed and deployed in the US. In 1978, the first satellite of the first Global Navigation Satellite System was launched, known as Global Positioning System (GPS) , and it started providing global coverage in 1994 [5].

Satellite based navigation systems providing global coverage, collectively known as Global Navigation Satellite Systems (GNSS), are used in many sectors due to passive positioning and timing capability. They can provide accuracy within a few centimeters in a short time. GNSS allows receivers on earth to calculate their position (longitude, latitude, and altitude/elevation) within a few meters using time signals transmitted along a line of sight by radio waves from satellites orbiting the earth. GPS was the first satellite-based navigation system developed by the U.S, other systems include GLONASS by Russian Federation, Galileo by European Union, BeiDou by China, Quasi-Zenith Satellite System (QZSS) by Japan, and the Indian Regional Navigation Satellite System (IRNSS). While GPS, Galileo, GLONASS, and Beidou are global navigation systems, QZSS and IRNSS are regional systems that can augment global systems and provide coverage to specific regions only.

GNSS work by transmitting signals from a constellation of satellites in space. These signals contain information about the satellite's location and the current time. GNSS receivers use this information to calculate their location on Earth. Since GPS is one of the oldest GNSS, it is fully developed and a lot of literature can be found related to it therefore, in this thesis it will be kept as the focus. However, other systems are similar to GPS and have changes in message structure as well as some additional features e.g., the short message service of BeiDou. Having that said, the positioning, timing, and navigation concepts are essentially the same for all the GNSS.

1.1 GNSS Segments

There are three main segments of GNSS systems, which include:

1. Space Segment (SS)
2. Control Segment (CS)
3. User Segment (US)

Figure 1.1 represent all three segments of GNSS

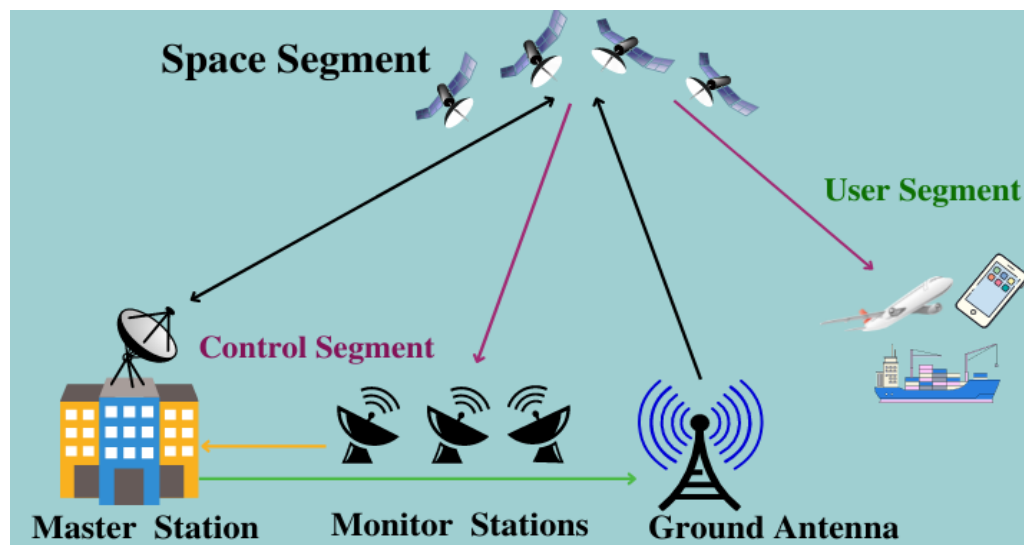


Figure 1.1 GNSS Segments

1.1.1 Space Segment

Space Segment refers to a constellation of satellites in Medium Earth Orbit (MEO), Low Earth Orbit (LEO), or geostationary orbit. Each satellite has its onboard atomic clocks to provide precise timing information for the signals it transmits. All satellites in the space segments are configured in orbits in such a way to ensure maximum visibility at any point on earth at any instant of time.

In the case of GPS, there are 32 satellites in the space segment. These satellites are distributed in 6 different orbits and each orbit contains 4 satellites. These orbits have a 55-degree inclination with respect to the equator and such a configuration provides visibility of 8 - 12 satellites for half a sidereal day (i.e., 11 58 minutes and 2 seconds). Moreover, these orbits exhibit a nearly circular shape with an eccentricity of less than 0.02, a semi-major axis of 26,560 Km, and an altitude of 20,200 km [6], [7].

GLONASS consist of 24 satellites in space segments are 24, distributed in 3 orbital planes whose ascending nodes are 120° apart and each orbital contains 8 satellites which are equally spaced with the arguments of latitude displacement of 45° . In the orbits, the satellites are placed at an altitude of 19100-km, incline 64.8° , and complete the orbit in the circular in approximately 11 hours 15 minutes. With the satellite spacing, the near-earth space and the terrestrial surface can be covered continuously and globally[3], [8].

European's GNSS system Galileo is composed of 30 satellites in Medium Earth Orbit, occupying three orbital planes inclined at 56 degrees to the equator with a 23,222 km altitude. Moreover, satellites orbit the earth for approximately 14 hours and spread evenly

in each orbit. In the Galileo constellation, 24 satellites are active and 6 satellites will be spare and used if any of the other satellites fail [9].

BeiDou china's GNSS system provide initial operational services in 2011 and the constellation is composed of 35 satellites, which are distributed as five in Geostationary orbits(GEO), three in inclined geosynchronous orbit(IGSO), and 27 in medium earth orbit(MEO) with 55-degree inclination to the equator with the latitude of 42164 km of semi-major axis in case of GEO and IGSO and 27878 km for MEO. Satellites are distributed in seven orbits, 1 for GEO, 3 orbits for IGSO, and 3 orbits for MEO[10].

Regional navigation system QZSS which is the Japanese System also known as Michibiki composed of four satellites. One satellite is in geostationary orbit and three are in Highly Elliptical Orbit (HEO). From Asia-Oceania, they are visible at all times because they will pass over the same ground track at the same altitude. The perigee altitude is 32000 km, and the apogee altitude is 40000 km. Among the three satellites of the QZSS, at least one is positioned near the zenith over Japan. Satellites appear almost overhead for more than 12 hours a day, with 70° plus elevations[11].

The other regional navigation system is IRNSS, which provides regional services about 1500km around India independently and autonomously. IRNSS space segment is composed of 7 satellites, three of which are geostationary and four are geosynchronous satellites in orbits of 250km perigee and 24000km apogee with 29° inclination[12].

Table 1.1 shows the specification of the GNSS constellation

Table 1.1 Specification of GNSS constellations

System	GPS 	GLONASS 	BeiDou 	Galileo 
Orbit	MEO	MEO	MEO, IGSO,GEO	MEO
Total Satellite	24	24	27,3,5	30
Constellation	6planes 56° inclination	Walker (24/3/1) 64.8° inclination	Walker (24/3/1) 55° inclination	Walker (24/3/1) 56° inclination
Services	SPS, PPS	SPS, PPS	OS, AS, WADS	OS, CS, PRS
Initial Services	Dec 1993	Sep 1993	Dec 2012	2016
Origin	USA	Russia	China	Europe
Coverage	Global	Global	Global	Global
Frequency (MHz)	L1 1575.42 L2 1227.60 L5 1176.45	L1 1602.00 L2 1246.00 L3 1202.025	B1 1561.098 B2 1207.14 B3 1268.52	E1 1575.42 E5a 1176.46 E5b 1207.14 E6 1278.75

1.1.2 Control Segment

Control Segment is the ground infrastructure used to monitor and control the satellites constellation. It consists of a number of monitoring stations and a master control station[3]. Monitoring stations track the satellites and calculate their orbits and the master control station is responsible for commanding and controlling the satellites. S-Band antennas are

used for two-way communications between the satellites and the control stations. The main tasks of the ground control station include:

- Generating precise navigation messages
- Command and control of constellation
- Upkeep of satellite clocks
- Orbit determination

The master control station of GPS is located at Air Force Base in Colorado Springs, United States. It contains atomic clocks that are used for time synchronization and for generating navigation messages (i.e., precise ephemeris). The navigation messages are transmitted to the satellites via S-Band antennas from each of the six ground monitoring stations.

1.1.3 User Segment

The User Segment is the equipment and personnel that use the GNSS signals to determine their position, velocity, and time. The user equipment can be portable, mobile, or fixed. The receivers used in the User Segment are typically much smaller and less expensive than the ones used in the Space and Control Segments.

In the early days, GNSS receivers were exceptionally large, expensive, and used to consume more power. But with the advancement in technology, GNSS receivers have become smaller in size, less expensive, and low power consuming. These portable GNSS receivers can be carried by a person and used for pedestrian navigation. The mobile GNSS receivers are installed in vehicles and used for vehicle navigation and fixed GNSS receivers are used for surveying applications.

1.2 GNSS Positioning Principle

In GNSS systems minimum of four satellites are required to determine the user position. Satellites transmit the signals and the GNSS system measure the distance between the satellites and the receiver. With these measurements, the user can calculate four unknowns and three unknowns are X, Y, Z coordinates or user position and last one is the clock error of the receiver[13], [14]. The principle of GNSS-based positioning is seen in figure 1.2.

Several factors influence the accuracy of distance measurements. Utilizing GNSS reference stations at known positions can help improve the accuracy of the user's position by quantifying the effect of errors on distance measurements. In positioning services, the modeled error size is provided as a correction to the user. This correction can help improve the positioning accuracy.

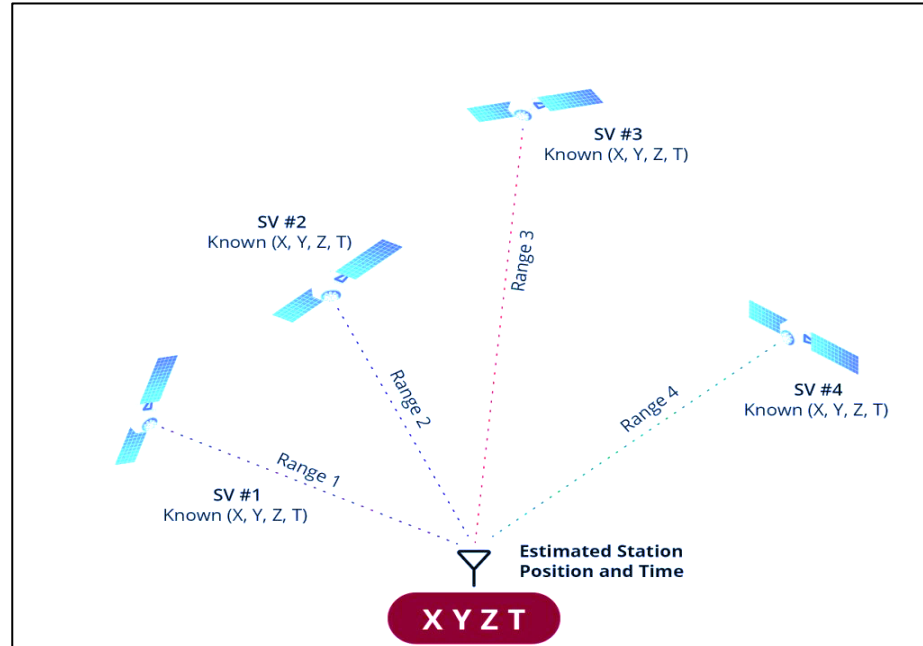


Figure 1.2 Positioning Principle: At least Four Satellites are required to estimate four unknown [15]

1.3 GNSS Signals

GNSS signals consist of three major parts which include:

- Carrier frequency (i.e., L-band L1, L2, & L5)
- Navigation message
- Pseudo-Random Noise (PRN) code

GPS uses 1575.42 MHz carrier frequency for its L1 while 1227.6 MHz & 1176.46 MHz carrier frequencies are used for L2 & L5, respectively. Apart from that, the Navigation data is transmitted at 50 bits per second while the code frequency is 1.203 MHz in case of coarse acquisition (C/A) and 10.23 MHz in case of precision P(Y) tracking.

GPS uses Code Division Multiple Access (CDMA) technique for multiple access. In this technique, each satellite uses a different PRN code (i.e., C/A or P(Y)) to modulate the carrier signal. This unique code helps in identifying the satellite as well as provides anti-spoofing protection to military users (in the case of L2 encrypted signals). First, the low data rate message is spread by a fast-chipping rate spreading sequence through a direct spread spectrum sequence. After that, it is modulated over the carrier. GPS uses Binary Phase Shift Keying (BPSK) modulation scheme which transmits navigation data in two phases i.e., 0° & 180° .

GPS signals are very weak (about -160 dBW) when they reach the earth's surface due to their long path length from satellites to GNSS receivers. Moreover, these signals are also attenuated by the ionosphere and troposphere. Therefore, GPS receivers use very sensitive receivers with various correction models to detect these weak signals and apply corrections to cater to several effects such as ionosphere, troposphere, and relativity.

1.3.1 GNSS Message Structure

The message structure of GPS signals is designed in such a way that both military and civilian users can use it. The message structure of L1 C/A & P(Y) signals consists of 5 sub-frames each lasting for 6 seconds. Each sub-frame further consists of 10 words. Thus, each GPS navigation message lasts for 30 seconds.

The first word of each sub-frame is called telemetry (TLM) which contains a fixed preamble[16]. It consists of a known bit sequence (10001011) that helps the GPS receiver to track the signal. The second word of each sub-frame is called the hand-over word that contains the truncated version of the z-count modulo-256. Followed by data bits and there are 6 parity bits in each word that tells whether the nav data is valid or not. First 3 sub-frames of the GPS message contain the ephemeris data which is the fresh orbital parameters along with the satellite vehicle (SV) health, information related to selective availability, and anti-spoofing. Ephemeris is updated every 2 hours. The last 2 sub-frames of the GPS message contain almanac data. Almanac is the least precise orbital information that repeats after 1 week. It helps speed up the acquisition of the signal and can be helpful in various applications.

Figure 1.3 provides a brief overview of the GPS message.

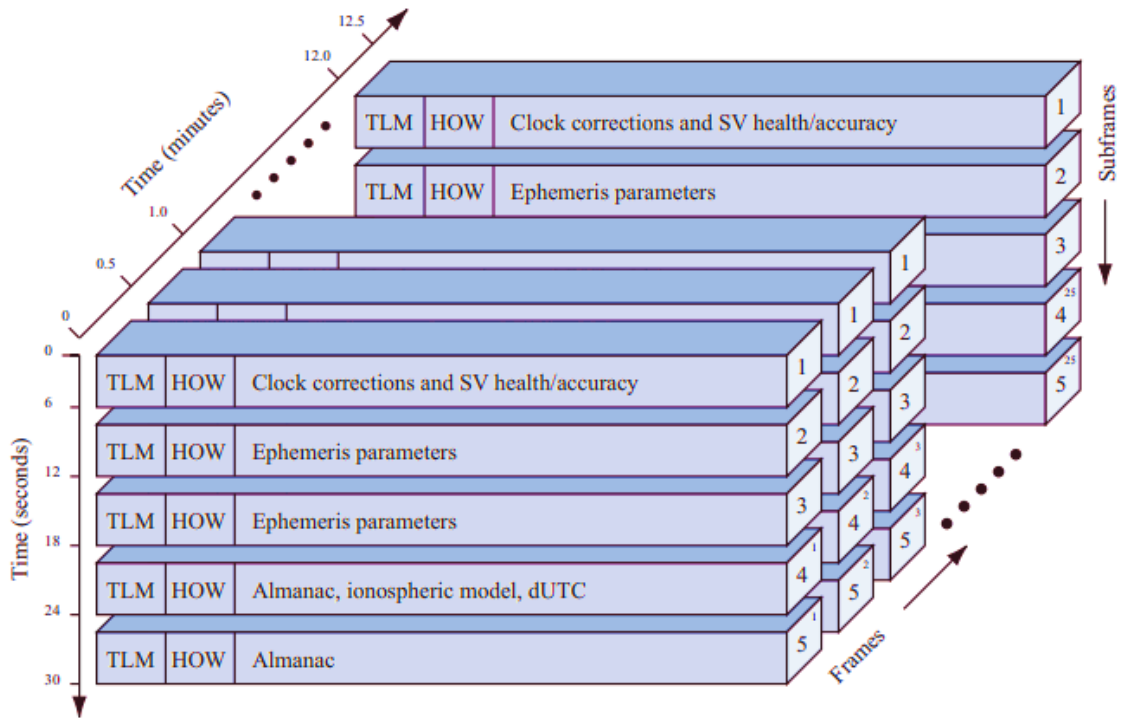


Figure 1.3 Message Structure [16]

1.3.2 Ranging Codes (C/A)

As mentioned, each GNSS satellite transmits its unique ranging code that helps in the identification of the satellite. There are 2 types of ranging codes used by GNSS satellites: coarse/acquisition (C/A) code & precision (P(Y)) code.

The C/A code is a pseudo-random noise (PRN) code that is modulated on the L1 carrier signal. The length of this code is 1023 chips or bits that repeat every millisecond. This code is available for public and civilian use. Furthermore, it can be generated using two linear feedback shift registers (LFSRs).

During the LSFR, G1 and G2 registers with the initial state all ones are shifted right by one bit. The output of G2 is an exclusive-or (XOR) with the output of G1. The first register i.e., G1 uses the following polynomial order

$$f(x) = 1 + x^3 + 10^{10} \quad (1)$$

while for G2 the following order is used:

$$f(x) = 1 + x^2 + x^3 + x^6 + x^8 + x^9 + x^{10} \quad (2)$$

The P(Y) code is a PRN code that is modulated on both L1 & L2 carrier signals and is used by authorized military users. The length of this code is much longer i.e., 1023 chips or bits & takes about 7 days to repeat. It can only be generated using the same LSFR with two extra registers[16].

Table 1.2 represents the delay sequence for all GPS satellites along with the first bits for each SV.

Table 1.2 Code Sequence

PRN Number	Code phase selection G^2	Code delay chips
1	$2 \oplus 6$	5
2	$3 \oplus 7$	6
3	$4 \oplus 8$	7
4	$5 \oplus 9$	8
5	$1 \oplus 9$	17
6	$2 \oplus 10$	18
7	$1 \oplus 8$	139
8	$2 \oplus 9$	140
9	$3 \oplus 10$	141
10	$2 \oplus 3$	251
11	$3 \oplus 4$	252
12	$5 \oplus 6$	254

13	$6 \oplus 7$	255
14	$7 \oplus 8$	256
15	$8 \oplus 9$	257
16	$9 \oplus 10$	258
17	$1 \oplus 4$	469
18	$2 \oplus 5$	470
19	$3 \oplus 6$	471
20	$4 \oplus 7$	472
21	$5 \oplus 8$	473
22	$6 \oplus 9$	474
23	$1 \oplus 3$	509
24	$4 \oplus 6$	512
25	$5 \oplus 7$	513
26	$6 \oplus 8$	514
27	$7 \oplus 9$	515
28	$8 \oplus 10$	516
29	$1 \oplus 6$	859
30	$2 \oplus 7$	860
31	$3 \oplus 8$	861
32	$4 \oplus 9$	862

1.4 GNSS Error Sources

The following are the main error sources that contributes to the overall signal of GNSS and dilutes its positioning performance. These errors are divided into three categories, starting from the space segment errors, followed by the propagation channel errors and ground segment errors. All these errors have been discussed below:

1.4.1 Satellite Clock Error

Each satellite is equipped with a highly stable atomic clock that controls the satellite timing and also helps in the generation of the signal. However, even the stable clock can experience drifts and as a result, up to 1 ms of deviation from the reference clock can occur. A master control station monitors the signal from each satellite and predicts the clock deviation based on the previous measurements. The measured clock biases and drifts are sent back to the satellite which are rebroadcasted to the user in the message signal. A complete navigation message structure can be seen in the later section of this thesis.

The broadcast clock error model is as follow:

$$dt^i = a_{f0}^i + a_{f1}^i(t - t_{oc}) + a_{f2}^i(t - t_{oc}) + \Delta t_r \quad (3)$$

Here $a_{f0}^i, a_{f1}^i, a_{f2}^i$ are the clock bias, clock drift and the frequency drift respectively, whereas t_{oc} the clock time at which the signal was received by the receiver and t is the current epoch time. Δt_r represents the correction of time due to relativity.

1.4.2 Ephemeris Error

An ephemeris error occurs whenever the broadcast ephemeris data in the message received from the satellite doesn't corresponds to the actual satellite position. Ephemeris obtained from the navigation message of each satellite signal are estimates by master control station based on the A-priory measured motion and position of satellite and the gravity of earth. There are three major components of the ephemeris error along with orthogonal directions that are defined relative to the satellite orbit:

- Along-track

- Cross-track
- Radial Error

Out of these three components the radial error directly affects the range measurements.

The ephemeris data is typically uploaded once a day by the command and control station due to which it stays fresh and as a result the root-mean-squared error is about 1.5 m [4].

1.4.3 Ionosphere and Troposphere Effects

The ionosphere is a layer of the earth's atmosphere that consists of electrons and ions. It lies from 60-1000 km above the earth's surface. It plays a very important role in the propagation of GNSS signals as it can refract these signals. The ionosphere is not homogeneous and its density changes with time, location, and season. This non-homogeneity of the ionosphere results in a delay in the GPS signals that can be up to 40ms. There are two main methods that are used to correct these delays, the broadcast ionosphere model & dual-frequency receivers.

The troposphere is the lowest layer of the atmosphere, and it extends from the earth's surface up to about 10 km. Figure 1.4 shows the propagation delays. The main component of the troposphere is water vapor which can cause a delay in the GPS signals. This delay is called wet delay and is mainly caused by the delay in the signal when it passes through the water vapor[17]. The wet delay can be up to 5ms. There are two main methods that are used to correct these delays, the broadcast tropospheric model & zenith total delay (ZTD).

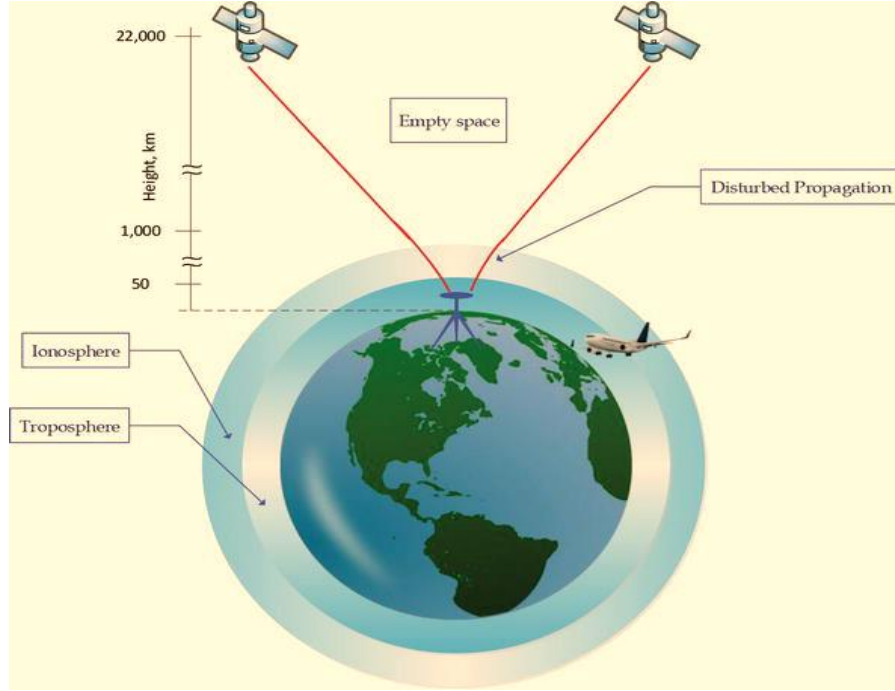


Figure 1.4 Propagation Delays [17]

1.4.4 Sagnac Effect

Besides the ionosphere and troposphere, there is another effect that can cause a delay in the GPS signals and this effect is called Sagnac effect. The Sagnac effect is caused by the rotation of the earth as figure 1.5 describes. It results in a delay that is proportional to the area enclosed by the antenna's beam. In navigation message ephemeris parameters provide satellite position in earth-centered earth fixed frame however, during the signal transmission due to rotation of the earth, there is a delay in transmission time and needs to mitigate. For the sagnac correction geometric range is measured as:

$$\rho_r^s = \|r_r(t_r) - R_z(\omega_e(t_r - t_t))r_s(t_t)\| \quad (4)$$

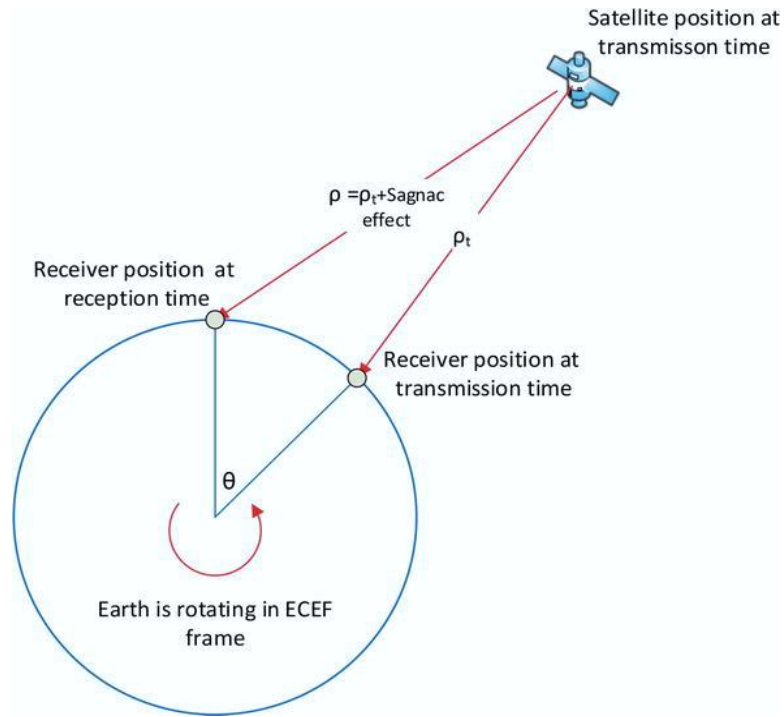


Figure 1.5 Sagnac Effect [17]

1.4.5 Multipath

Multipath is caused when the GPS signals reflect off of buildings or other objects before reaching the antenna and due to the motion of the receiver as shown in figure 1.6. The reflected signals from different surfaces can cause interference with the direct signals and results in a skewed correlation peak. Furthermore, the multipath is an uncorrelated error that causes range delays.

This can cause a delay of up to several microseconds. It affects both the carrier and code measurements. In comparison to carrier phase solutions, the effect on pseudorange solutions is orders of magnitude greater due to the fraction of a carrier wavelength represented by the multipath and the code phase associated with the chipping rate.

There are two main methods that are used to mitigate the effect of multipath i.e., by using a ground plan below the antenna and by placing the nulls of an antenna in the direction of the reflected signals.

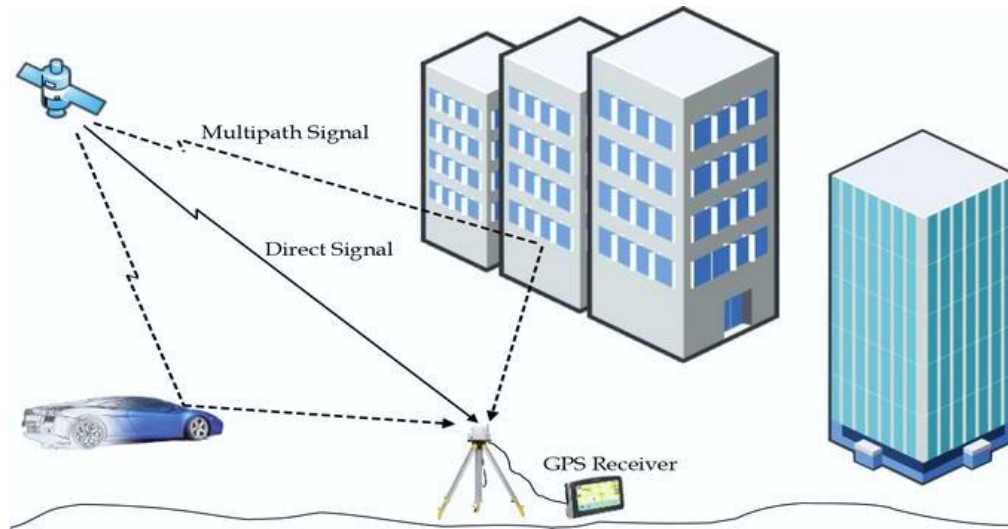


Figure 1.6 Multipath Error [17]

1.4.6 Receiver Clock Errors

GNSS satellites are equipped with a highly stable atomic clock that provides highly accurate timing. However, the receiver, on the other hand possesses a standard quartz crystal clock that is not very stable compared to the atomic clocks of the satellite. Consequently, the error in the receiver clock is much larger than the GNSS clock. This can highly affect the positioning performance of the system because GNSS is strictly based on the time of arrival method. To solve the problem of the receiver clock there are two main options. The first option is to use a highly stable external clock such as rubidium or cesium-based clocks. But the problem is these clocks are extremely expensive and thus, it will highly affect the price of a common user grade GNSS receiver. Another method is to use the timing information received from the satellites for the correction of the receiver clock.

This method is relatively more efficient and cost effective as it doesn't require the use of any external clock.

1.5 Doppler Effect

Another factor that affects the GNSS signals is the Doppler effect, due to the relative motion between the satellite and the receiver. It results in a frequency shift of the GPS signals. The amount of frequency shift is proportional to the relative velocity between the satellite and the receiver. Due to the motion of the satellite and receiver, the maximum Doppler shift can go up to ± 10 kHz in case of the carrier frequency while in case of the code the Doppler shift can be up to ± 6.4 Hz at max. This frequency shift is searched by the receiver during the acquisition phase.

1.6 Dilution of Precision

In satellite navigation and geomatics engineering, dilution of precision (DOP) or geometric dilution of precision (GDOP) describes the error propagation resulting from the geometry of navigation satellites on the solution precision. The Receiver Autonomous Integrity Monitoring (RAIM) algorithm depends on satellite geometry and requires a minimum of 6 satellites for the Fault Detection and Exclusion (FDE) technique with good satellite geometry[18],[19]. In figure 1.7 geometry of bad and good DOP is well explained.

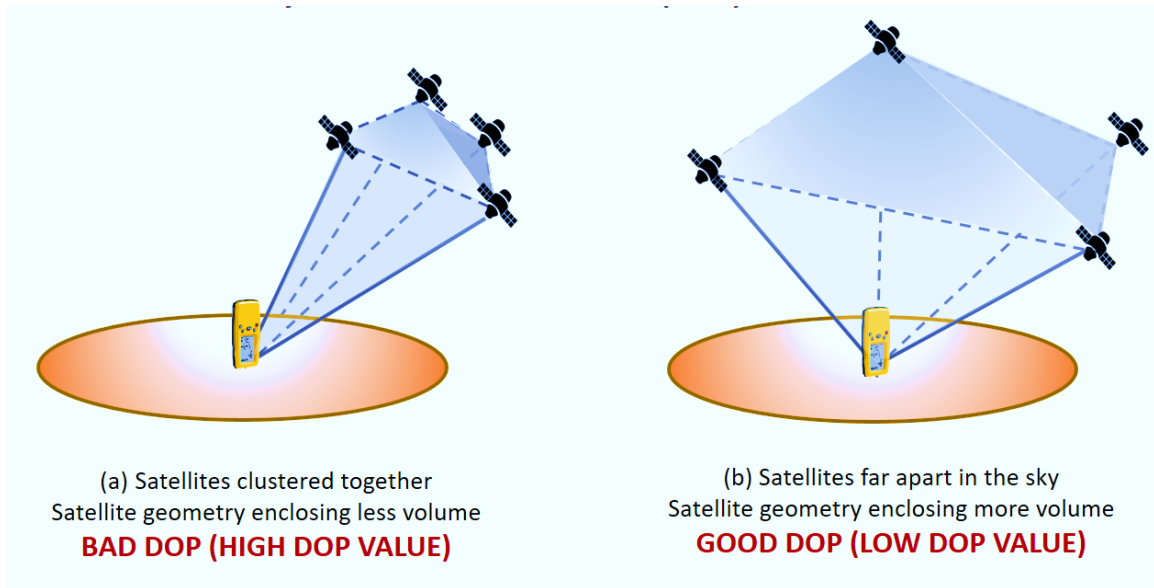


Figure 1.7 Satellite Geometry and DOP [19]

1.7 GNSS Performance Parameters

In a navigation system, the following basic four parameters are required to characterize the performance of navigation [20].

i) Availability

Availability of the system is the total time for which the solution of the navigation system is available to the user. Availability of the system depends on the satellite constellation, and it varies considerably depending on the type of application and services used, but usually ranges from 95 to 99.9% [21].

ii) Continuity

Continuity is the ability of the system to provide uninterrupted navigation solution with steadfast accuracy throughout the time. In safety-critical applications, continuity plays an

important role as it provides the required performance of the system during the entire phase of the mission.

iii) Integrity

Integrity refers to the measure of trust that can be placed in the correctness of the position or time estimate provided by the receiver. It also provides a timely warning to the system when the provided solution is not able to trust.

iv) Accuracy

Accuracy provides the difference between the calculated or estimated position or time and the actual solution of the navigation system. It is meaningless in terms of navigation system if it does not include the statement of uncertainty.

Among all these four parameters integrity of the system is the most important to monitor because it ensures whether the solution provided by the system can be trusted or not and it can be monitored at a user level. It is used for safety-critical applications such as aircraft, ground-based autonomous vehicles, fleet management, autonomous rovers, etc. Besides that, the availability and accuracy of the system generally depend upon the receiver's capability. If a receiver is able to receive signals from multiple constellations, then the availability and accuracy of the output will be much higher and vice versa.

1.8 Problem Statement

GNSS code measurements suffer from integrity issues which is detrimental for safety-critical applications. The integrity monitoring methods such as Ground Based Augmentation System (GBAS), Satellite Based Augmentation System (SBAS), and

Ground Integrity Channel (GIC) are costly and not available for common user applications as they require a subscription. RAIM algorithms mitigate this issue by implementing integrity monitoring at user end. However, each algorithm has its downside. One or more algorithms can be integrated, complementing each other, thus forming an effective integrity resolution scheme.

1.9 Motivation

Single frequency receivers are cost-effective and widely available, but they don't have high positioning accuracy, on the other hand dual-frequency receivers are accurate but expensive. The positioning accuracy provided by low-cost single frequency receivers in various situations can be enhanced by timely generating warnings and excluding the faults whenever needed by implementing integrity monitoring algorithms. To complement integrity monitoring issues, an improved method based on a polling scheme of least squared residuals testing and correlation analysis method is devised for single frequency code measurements.

2. Literature Review and Background

In GNSS different systems are used for integrity monitoring like GNSS Integrity Channel (GIC), Satellite Based Augmentation System (SBAS), Ground Based Augmentation System (GBAS), and Receiver Autonomous Integrity Monitoring (RAIM). Currently, GPS provides integrity information in navigation messages but it's not timely enough for safety critical applications as it's not reliable. The taxonomy of different integrity monitoring schemes can be seen in figure 2.1.

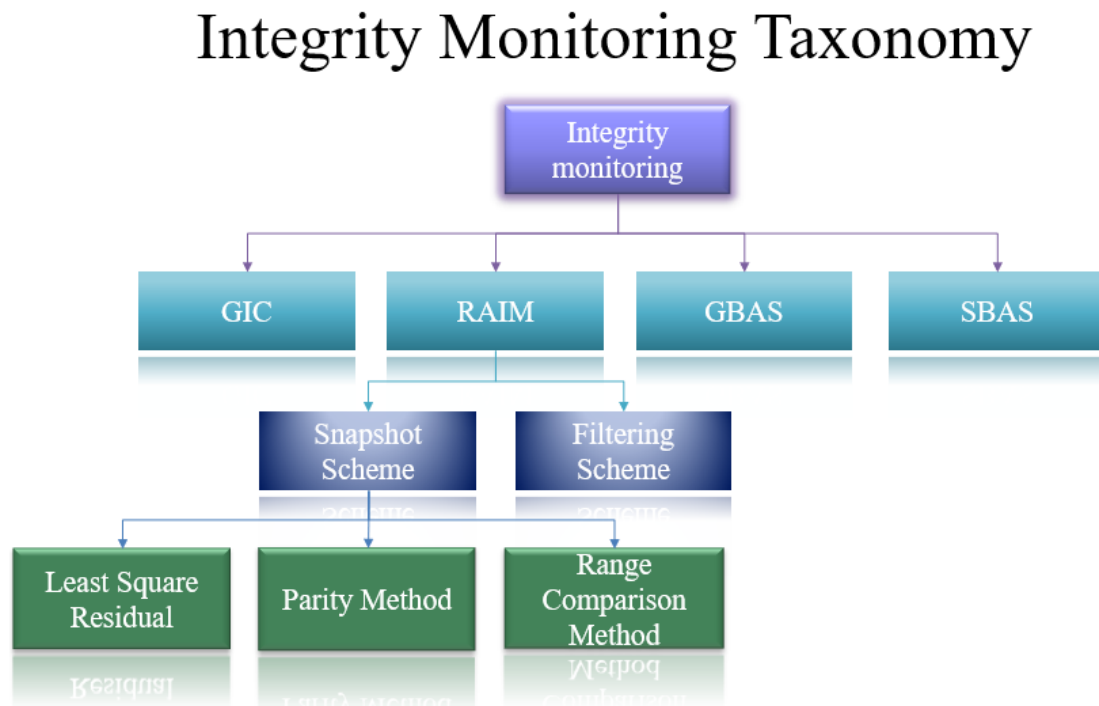


Figure 2.1 Integrity Monitoring Taxonomy

In this research Receiver Autonomous Integrity Monitoring (RAIM) algorithm is used as GIC, SBAS, and GBAS depends on the ground monitoring stations and required a mean of conveying the information to the user and have costly infrastructure[22]. Apart from the time delay and cost GIC cannot provide complete integrity information as there are local

biases like multipath, interference and it cannot detect the biases[23]. While RAIM is cost efficient and based on current redundant measurements and doesn't require any external mean. It is self-contained or more simply it is a receiver built in algorithm and easy to monitor at a user level.

2.1 Navigation Solution Determination

Once the tracking loop extracts the navigation data bits from the signal after the removal of code and carrier wave. The ephemeris parameters are recovered from the signal which is then used for the computation of satellite position, time of transmission, and pseudorange between the user and the satellite. These values are the components of the user position calculation. There are several methods for the calculation of user position from received signal data but the most widely used methods are least square estimate and the weighted least square estimate.

In order to determine the user position in 3 dimensions as well as the receiver clock offset, pseudorange needs to be calculated for at least 4 satellites. Let ρ represents the pseudorange between the user and the satellite then the equation becomes:

$$\rho_s = \|x_s - x_u\| + ct + \varepsilon_{\rho_s} \quad (5)$$

In equation (5) x_s represents the known satellite position while the x_u represents the unknown user position. C is the speed of light which is 3×10^8 m/sec and t is the user clock bias. ε_{ρ} is the combined error including tropospheric delays, ionosphere delays, ephemeris estimation errors and noise of the receiver. To extract the user position coordinates from the equation (5) it needs to linearize first about a nominal value. Usually these are the best

estimate [24]. Using the state and bias estimates the predicted pseudorange measurements can be obtained as follow

$$\rho_{s,0} = \|x_s - x_0\| + ct_0 + \varepsilon_{\rho_s,0} \quad (6)$$

The pseudorange measurement vector can be reduced as follow:

$$\Delta\rho = \rho_0 - \rho \quad (7)$$

Here in equation (7) ρ and ρ_0 are the predicted pseudorange measurement and the actual values respectively. The pseudorange equation can be linearized by using the Taylor expansion as follow:

$$\Delta\rho = A\Delta x + \Delta\varepsilon_\rho \quad (8)$$

where A is geometry matrix consist of direction cosine vector whose fourth column is 1, which represent the receiver clock state. While Δx is vector represents the offset of the user's true position and time bias, relative to linearization point and υ is the measurements error vector containing both deterministic and random errors. After obtaining the unknowns, user positioning coordinates x, y, z , and clock bias are acquired as

$$x = x_0 + \Delta x \quad (9)$$

2.1.1 Least Squares Navigation Solution

Observations are degraded by the errors, to reduce the effect of error, a larger number of measurements are required than the unknowns. When the measurements are larger than four, an overdetermined linear system of equations has to be solved as a result. In order to examine the problem, let's assume $m \times 1$ measurement vector y related to unknown vector x using the following linear relationship

$$y = Ax + v \quad (10)$$

A represents the satellite to user direction cosine vector of $m \times 4$ whose fourth column is 1, which represent the receiver clock state and v is the $m \times 1$ measurements error vector containing both deterministic and random errors. Furthermore, the linearized model assumes the expected value and variance as

$$\begin{aligned} E(y) &= Ax \\ D(y) &= \sigma_0^2 \end{aligned} \quad (11)$$

Least square minimizes the sum of squared residuals i.e.,

$$\min_x \|Ax - y\|^2 \quad (12)$$

If the quantity is set to zero, the following result will be obtained:

$$\hat{x} = (A'A)^{-1} A'y \quad (13)$$

Equation (8) is referred as least square estimate.

2.1.2 Weighted Least Square Navigation Solution

The weighted least squares (WLS) estimate is equivalent to the best linear unbiased estimate if the data's inverse variance covariance matrix is utilized as the weight matrix. By minimizing the total of the weighted discrepancies between the observations, weighted least squares attempts to find the optimal answer.

$$\hat{x} = (A'Q_y^{-1}A)^{-1} A'Q_y^{-1}y \quad (14)$$

where Q_y is the size $m \times m$ Variance Covariance Vector (VCV) matrix for observations and Q_y^{-1} is the cofactor or weight matrix for observations

2.2 Receiver Autonomous Integrity Monitoring (RAIM)

In an urban scenario, the navigation solution is degraded due to gross errors, and we need to check the reliability and integrity of the solution so current redundant measurements need to check and provide timely warnings to the user. Receiver Autonomous Integrity Monitoring (RAIM) provides a timely warning when the presence of anomaly is detected.

In RAIM two schemes are used for the integrity monitoring: Snapshot Scheme and Filtering Scheme [25].

2.2.1 Filtering Scheme

In filtering scheme both current and a priori assumptions are used for error detection and identification. It is recursive algorithm that uses past measurement and update current epoch by propagating to the present measurement. Due to which filtering scheme is computationally complex and sometimes false a priori assumptions cannot provide reliable results. However, even with good measurements, a poor dynamic model will result in a suboptimal solution. For navigation applications, filtering schemes are often preferred. In severe signal degrading conditions, filtering scheme is not able to offer the usual advantages due to the absence of additional sensor, and the limited knowledge about user dynamics [26].

2.2.2 Snapshot Scheme

In the Snapshot scheme, only current measurements are used and decision as failure or no-failure is based on it. It gained more acceptance as it is not computationally complex. Snapshot scheme is further classified as a least squared residual method, parity method and

range comparison method. All these methods based on noisy redundant measurements and the basic measurement equation is common to all the RAIM methods.

$$y = Ax + v \quad (15)$$

2.2.2.1 Least square residual method

In the least square residual (LSR) method, a residual vector $\hat{v}$ is calculated to measure the consistency of measurements based on least squares adjustment. The squares sum of the pseudorange residual plays a vital role as the basic observable in the LSR and the test statistics (SSE) are [27]

$$SSE = \hat{v}^T \hat{v} \quad (16)$$

It is the scalar and non-negative quantity, which is used for detection of erroneous measurements. The further details of least square residual method are in section 2.4.

2.2.2.2 Parity method

In parity method observation vector is transformed into parity space vector. The erroneous measurement is then detected by testing the abrupt changes of the observation error and extracted. The linear transformation is performed as

$$p = Py \quad (17)$$

where P is parity matrix $(n-4) \times n$ and obtained by QR factorization of geometry matrix A , whose rows are mutually orthogonal to the column of the A matrix, and unity in magnitude. The sum of the squared residuals (SSE) is the identical in both range space and parity space. In parity method the test statistics (SSE) is

$$SSE = p' p \quad (18)$$

In detection application, the least-squares-residuals and parity methods lead to identical observables, and it is the important function in safety critical applications.

2.2.2.3 Range comparison method

For implementing range comparison, more than four satellites must be present. In this case, we would have equations with four unknowns. The calculation of the remaining measurements is based on a solution that satisfies the equations. In order to compare estimated values with actual measurements, estimated values are used. In the case of small residual differences, the detection algorithm declares "no failure", and in the case of large residual differences, it declares "failure"[24] .

In this research, the least squared residual method is as it is more efficient and easier to implement. The system is also capable of accurately deriving instantaneous position fixes at corresponding epochs by detecting rapidly growing errors with no reliance on past measurements. This research aims to develop a reliable algorithm for low-cost single frequency GNSS receivers that currently hold more than 60% of the market.

Traditionally, GNSS based measurements have been enhanced with the help of RAIM for the aviation sector to meet the International Civil Aviation Organization (ICAO) standards. Different studies on the identification of outliers are available in the literature since 1980 such as [28]. In 1986, Y. Lee [29] devised the range comparison and position comparison method. In this research, the author proved that the two methods are mathematically equivalent. However, the results from this study provide evidence that comparison methods alone are not sufficient. But when combined with the user clock bias estimate monitoring, it has some potential to detect the user position error. The author in [30] further extended

the receiver autonomous integrity monitoring by using the parity method using the current measurements.

During this study, the probability of false alarm and missed detection were related to each other along with the measurement observation matrix and the ratio of detectable bias shift to standard deviation of measurement noise. In addition to this, another study can be found in the literature from the same year in which the author presented a new method for satellite failure detection and isolation. The range residual is used as test statistics for the algorithm presented in this study [31]. The range comparison method is further improved for standalone navigation [32] and different baseline RAIM methods are explained in [27] in 1992. The author of [33] assessed the performance levels of GNSS RAIM along with reliability measures and in [34] implementation of improved fault detection and exclusion algorithm using parity method is explained. Furthermore, the advancements in the RAIM algorithms are mentioned in [35]–[38]. The RAIM model present in this research is based on four tests for detection and identification including global test, LSR test, correlation coefficient, and correlation distance.

2.3 Fault Detection and Exclusion

RAIM operations fall into two main categories. The first one is fault detection and exclusion (FDE) while the next one is position and protection levels calculation. Fault detection can be done using 5 satellites only. However, for the detection and exclusion, a minimum of 6 satellites are required. FDE excludes the faulty measurements and then allows the system to continue to ensure the overall integrity[39], [40]. It takes standard deviations, measurement geometry, fixed probabilities of false alarms (P_{FA}), and missed detection (P_{MD}) as input. Based on this input it generates either fault detection or protection

levels (horizontal or vertical). That will guarantee that the true position is present in the circle with a given probability of P_{FA} and P_{MD} . The fault detection and exclusion method requires a decision threshold which is a critical value for its computation. This decision threshold can be calculated from the probability of false alarm and missed detection [34].

The navigation system error should be controlled within a certain range. This range is called protection level(PL). PL can be classified as vertical protection level (VPL) and horizontal protection level (HPL). These protection levels are decided by the positioning accuracy of the GNSS. In addition to this HPL and VPL are inversely proportional to the accuracy which means greater the accuracy of the system the smaller will be its protection level. Now whenever a navigation system is at fault its navigation accuracy will reduce under such conditions the measurements will exceed the protection levels and a warning will be generated [41].

RAIM depends on the geometry of the visible GNSS SVs. To carry out perfect calculations and allow the algorithm to work flawlessly, the system needs to have sufficient SVs in orbit too. As mentioned earlier 5 SVs are needed for fault detection and 6 SVs for exclusion. However, this is the minimum requirement and isn't sufficient. The final subset formed by eliminating one SV at a time must have sufficient geometry for fault detection capability.

2.4 RAIM Algorithm

2.4.1 Methodology

The proposed RAIM scheme is based on fault detection and exclusion principle using weighted least square algorithm. The algorithm is based on linearized Gauss – Markov model of geodetic adjustment for m number of known measurements and n number of

unknowns [42], [43]. For example, if we have m number of available satellites then the equation can be formulated as follow:

$$y = Ax + v \quad (19)$$

y represents $m \times 1$ vector of observed minus computed pseudo range also known as measurement vector, A represents the satellite to user direction cosine vector of $m \times 4$ whose fourth column is 1, which represent the receiver clock state. While x is 4×1 vector represents the unknowns containing user positioning coordinates x , y , z and clock bias and v is the $m \times 1$ measurements error vector containing both deterministic and random errors. The above equation is solved using least square which is given as:

$$\hat{x} = (A'Q_y^{-1}A)^{-1}A'Q_y^{-1}y \quad (20)$$

In equation (2) Q_y size $m \times m$ represents the diagonal weight matrix whose elements are the reciprocal of variance of each measurement [44]. The least square adjustment yields measurement residual which is as under

$$\hat{v} = y - A\hat{x} \quad (21)$$

The variance covariance matrix of the residual can be calculated using Gauss error propagation law which results an $m \times m$ matrix.

$$\hat{v} = y - A(A'Q_y^{-1}A)^{-1}A'Q_y^{-1}y = (I_n - A(A'Q_y^{-1}A)^{-1}A'Q_y^{-1})y = Q_{vv}y \quad (22)$$

$$\hat{v} = Q_{vv}\varepsilon \quad (23)$$

$$Q_{vv} = Q_y - A(A'Q_y^{-1}A)^{-1}A' \quad (24)$$

Q_{vv} is also known as sensitivity or mapping matrix. Outlier can be detected and identify by residual vector through the mapping Q_{vv} , as there is linear relationship between residual vector and measurement error vector.

2.4.2 Fault Detection Method

For fault detection global test is performed, the test statistic is formulated as weighted sum of squares of the least square residual divided by difference between the number of available satellites and number of unknowns also known as redundancy ($m - n$) [45].

$$SSE = \frac{\hat{\mathbf{v}}\mathbf{Q}_y^{-1}\hat{\mathbf{v}}}{m - n} \quad (25)$$

For integrity monitoring, fault detection is a key factor which is based on the hypothesis testing of global test. The general form of hypothesis tests can be expressed as follow:

$$\begin{cases} H_0: SSE \leq T \\ H_1: SSE > T \end{cases} \quad (26)$$

Fault detection is carried out by comparing test statistics and identification threshold. H_0 shows the null hypothesis when no fault is detected and H_1 shows the alternate hypothesis when fault is identified.

For testing inconsistency in measurement, the central Fisher-distribution $F_{1-\alpha}(m - n, \infty, 0)$ and Chi-square test $\chi_{1-\alpha, m-n}$ is used in RAIM, where α is the significance level i.e., false alarm rate [46]. To ensure the null hypothesis with respect to alternate hypothesis the global consistency test is used. If the test statistics is exceeding the threshold value, the erroneous measurement is detected, and null hypothesis is rejected. After the detection of erroneous measurement, outlier is identified and eradicated.

$$SSE > F_{1-\alpha}(m - n, \infty, 0) \quad (27)$$

2.4.3 Protection level

A GNSS receiver's reliability must always be determined by the amount of trust, applications are able to place in the system at any given time. In addition to improving the accuracy of the position information and keeping people and assets safe, the GNSS receiver calculates the protection level in real-time. A receiver's protection level describes the maximum possible error within a specified confidence level. It can be taken as threshold for fault detection. The protection level (PL) represents the radius of the circle, while the true position is located at the center of the circle[15],[28],[47]. Horizontal protection level (HPL) provides bounds to horizontal direction and vertical protection level (VPL) provides bounds to vertical position. HPL is determined as

$$HPL = HSlope_{max} + P_{bias} \quad (28)$$

$HSlope_{max}$ is the maximum horizontal slope, which represents the degree of sensitivity to bias in the horizontal position error for any pseudorange measurement and formulated as [27],[28]:

$$HSlope_i = \sqrt{(M_{1i}^2 + M_{2i}^2) / Q_{vv}} \quad (29)$$

where $M = (A'Q_y^{-1}A)^{-1}A'Q_y^{-1}$ shows that geometric matrix and HPL are highly correlated and Q_{vv} is the sensitivity matrix calculated as above in equation (6). P_{bias} is the minimal detectable bias determined by setting the threshold in the failure detection test, probability of missed detection and variance σ_o .

$$P_{bias} = \sqrt{\lambda} \times \sigma_o \quad (30)$$

where λ represents the non-centrality parameter of the chi-square density function. Protection levels for each application varies, because of the desired reliability of the GNSS receiver and of position estimate. Figure 2.2 shows the flow chart of fault detection method.

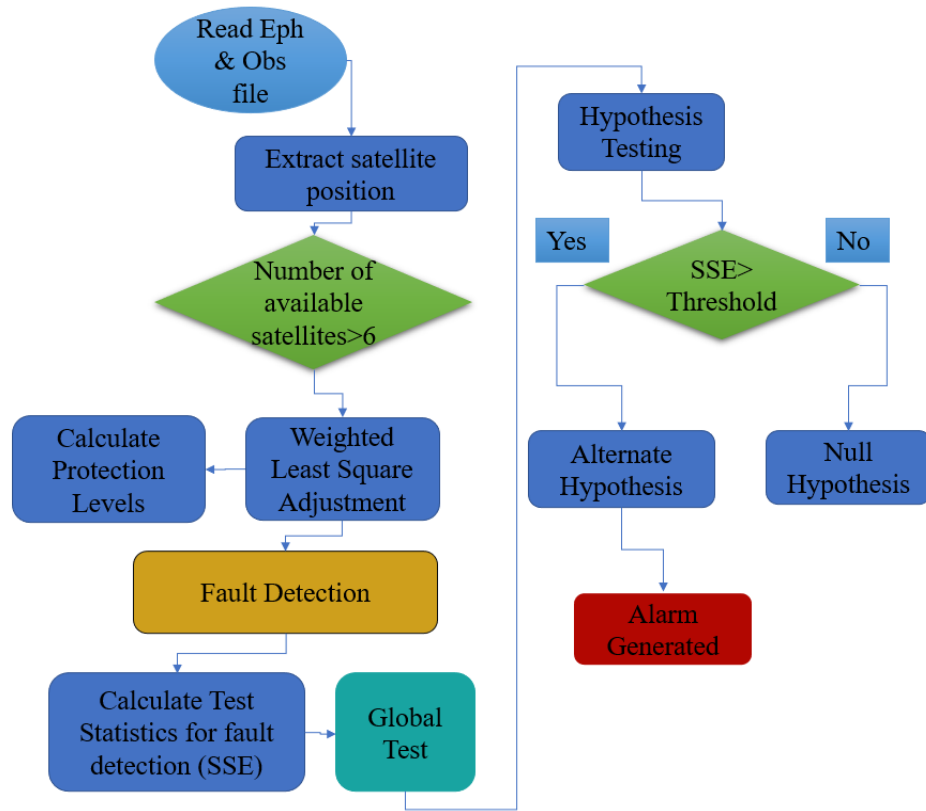


Figure 2.2 Fault Detection

2.4.4 Fault identification testing

2.4.4.1 Least Square Residual (LSR) Test

Once the error is detected through F -distribution and Chi-square test, identification and exclusion of error is necessary for reliable positioning. For fault identification the local test is based on LSR testing. So, if alternate hypothesis of global test is accepted, local test is performed for identification and exclusion of error. The error is assumed to be normally distributed. Test statistic is calculated for i -th measurement as:

$$w(i) = \frac{\mathbf{c}' \mathbf{Q}_y^{-1} \hat{\mathbf{v}}}{\sqrt{\mathbf{c}' \mathbf{Q}_y^{-1} \mathbf{Q}_{vw} \mathbf{Q}_y^{-1} \mathbf{c}}} \quad (31)$$

LSR test identifies error when:

$$|w(i)| > N_{\frac{\alpha_0}{2}}(0,1)$$

The i -th measurement is considered as an error and excluded from measurements. A single outlier in one observation tends to increase several values of $|w(i)|$. The largest measurement with value higher than threshold is considered as an outlier [49]. α_0 is the significance level of the local test which is predefined. Value of both parameters of significance level of global and local test is predefined, both the parameters are interrelated, along with the probability of missed detection β , which remains same for both tests.

In order to ensure an accurate fault detection, correlation analysis methods are used in addition to LRS test to detect and eliminate erroneous measurements.

2.4.4.2 Correlation analysis method

Correlation analysis is concerned with examining the relationship between two variables, or to put it more simply, measuring the degree of correspondence between two random variables [50]. In fault identification, correlation analysis is performed to identify the erroneous measurement by measuring the degree of correspondence between measurement residual vector $\hat{\mathbf{v}}$ and measurement error vector $\boldsymbol{\varepsilon}$, and the relationship between these vectors is :

$$\hat{\mathbf{x}} = \mathbf{Q}_{vv} \boldsymbol{\varepsilon} \quad (32)$$

In order to measure the degree of correlation, two factors are considered: correlation co-efficient and distance correlation. The correlation co-efficient $c(i)$ of $\mathbf{Q}_{\mathbf{v}\mathbf{v}}$ and $\hat{\mathbf{v}}$ is calculated by given formulae:

$$c(i) = \frac{\sum_{j=1}^m (\mathbf{Q}_{ji} - \bar{\mathbf{Q}}_i)(\hat{\mathbf{v}}_j - \bar{\hat{\mathbf{v}}})}{\sqrt{\sum_{j=1}^m (\mathbf{Q}_{ji} - \bar{\mathbf{Q}}_i)^2 \sum_{j=1}^m (\hat{\mathbf{v}}_j - \bar{\hat{\mathbf{v}}})^2}} \quad (33)$$

and distance correlation $d(i)$ is calculated as [51]:

$$d(i) = \sqrt{\frac{1}{m} \sum_{j=1}^m (\mathbf{Q}_{ji} - \hat{\mathbf{v}}_j)^2} \quad (34)$$

where $i = 1, 2, \dots, m$, $\mathbf{Q}_i$ is the transpose of i -th column of $\mathbf{Q}_{\mathbf{v}\mathbf{v}}$ and $\mathbf{Q}_{ji}$ is the j -th element of $\mathbf{Q}_i$. $\bar{\mathbf{Q}}_i$ and $\bar{\hat{\mathbf{v}}}$ are the mean of $\mathbf{Q}_i$ and $\hat{\mathbf{v}}$ respectively. Correlation analysis method implies the following two tests for identification of outlier. The highest value of absolute correlation co-efficient and the smallest value of correlation distance is considered as outlier[49].

After the fault detection and identification, we compare the results of all the identification tests. Identification tests: w-test, correlation-coefficient and distance correlation are named as k1, k2 and k3. A polling scheme is implemented such that if all the three tests identify the same outlier, identified outlier is exclude and recalculate the position. In figure 2.3 fault identification steps are explained.

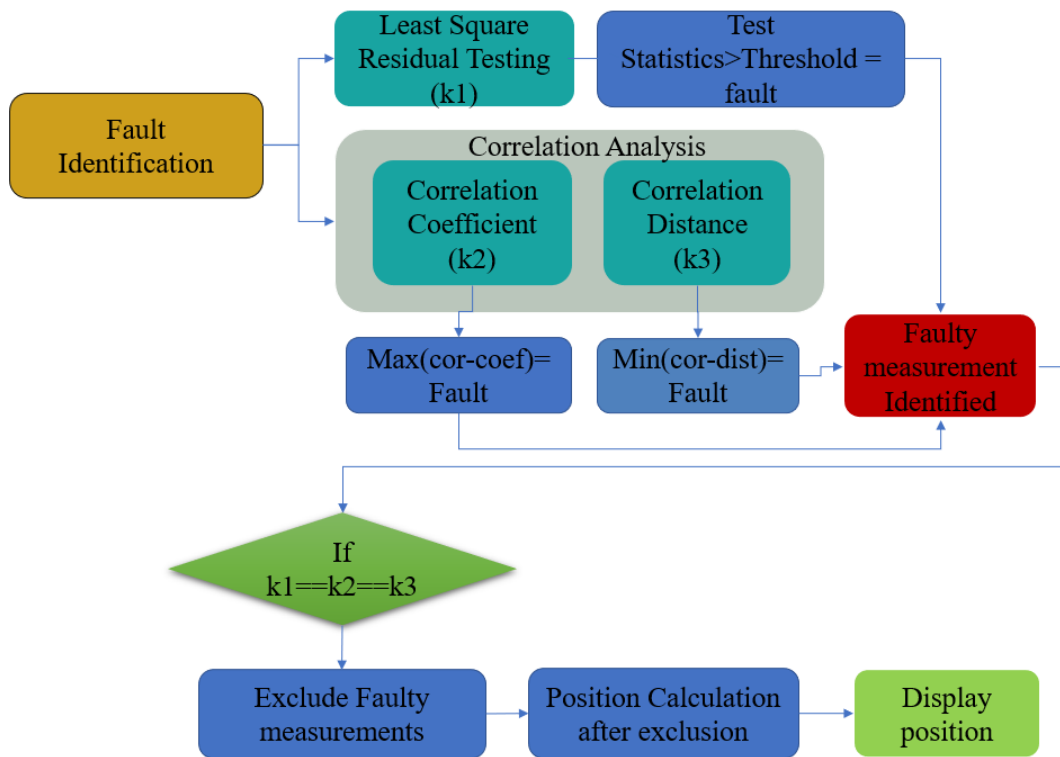


Figure 2.3 Fault Identification

3. Data Collection and Experimental Setup

3.1 Experimental Setup



Figure 3.1. Experimental setup for Dataset 1 & 2



Figure 3.2. Experimental setup for Dataset 3

3.2 Data Collection

For data collection, two receivers Emlid Reach RTK and Ublox C94 M8P were used which are single frequency dual constellation GNSS receivers, as shown in figure 3.3. To control the interface of receiver ublox receiver Ucenter was used. The receiver was configured on single positioning mode with an elevation mask of 10 degrees to maintain a good satellite geometry and reduce the effect of multipath from the surrounding. The sampling frequency was set to 1 Hz and only GPS satellites were selected during the data collection.



Figure 3.3. GNSS Receiver

To collect the data two different sites in different scenarios and time were selected for both dynamic and static datasets. Dynamic datasets were collected under clear sky in the IST Khatana Stadium at the Institute of Space Technology (IST), Islamabad, Pakistan while the static data set was collected at the rooftop of academic block 7 at IST. We required only dynamic dataset and static dataset was used only for algorithm testing.

The details for both the datasets are given in Table 3.1, while the location and track of both datasets are shown in Fig. 3.4 respectively.

Datasets were collected in different formats, and we required them in Receiver Interchange Navigation Exchange format (RINEX). To convert the datasets into RINEX format two software Emlid Studio and RTKLib were used.



Figure 3.4 Data Collection Sites

During the collection of Datasets 1, 2, and 3 the sky plots of available satellites can be seen in figures 3.5, 3.6, and 3.7.

Table 3.1 Datasets Details

Location	IST khatana Stadium (Dataset-1)	IST khatana Stadium (Dataset-2)	IST Block-7 (Dataset -3)
Date	28 th -Feb-22	2 nd -June-22	15 th -Feb-22
Number of available Satellites	13,15,17,28,30,14,1 9,1,24	32,22,10,23,21,31,2 7,8,24,18,1	1,6,7,13,14,15 ,17,19,21,28, 30
Number of Epochs	600	287	473

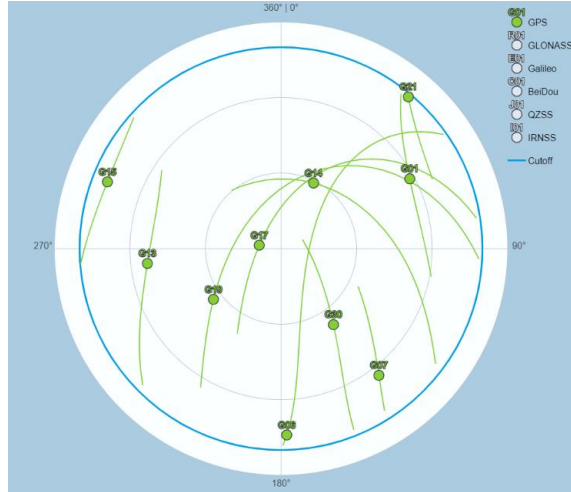


Figure 3.7 Satellite Visibility Sky Plot(Dataset-3)

3.3 Algorithm Implementation

For algorithm development, MATLAB is used as it is user friendly and easy to understand. MATLAB is a popular way to perform complex matrix calculations and produce output graphs in a friendly easy manner.

3.3.1 RINEX Files

Satellite navigation system raw data are stored in Receiver Independent Exchange format files based on a specified interval of time. These files help the users to correct their data in post-processing, improving its accuracy. As new types of measurements and satellite navigation systems emerge, RINEX's format is designed to evolve. For GNSS post-processing, RINEX Data Consist of the following file format:

- ❖ Data Navigation file
- ❖ Data Observation file
- ❖ Data Metrological file

In our research first two files are used in the algorithm.

Data Navigation file

RINEX Nav file [n] allows to read and write Broadcast Ephemeris data into and out of RINEX Navigation files. In this file, we will find Broadcast Ephemeris data conforming to RINEX standards in American Standard Code for Information Interchange (ASCII) format. Broadcast Ephemerides (also known as nav files) contain position, velocity, and clock information along with the auxiliary information such as health, satellite frequency, and age of information for GNSS constellations. All GNSS constellations use RINEX formats. And details are shown in figure 3.8.

1	2.10	N: GPS NAV DATA				RINEX VERSION / TYPE	
2	RTKCONV 2.4.2	20220301 052927 UTC				PGM / RUN BY / DATE	
3	log: E:\check\easy1-6\easy (1-6)\Field Data\RINEX\2nd rove					COMMENT	
4	format: RINEX					COMMENT	
5	PRN	Epoch/time	Ephemeris			END OF HEADER	
6	17	22 2 28 9 59 44.0	.581009779125E-03	.511590769747E-11	.000000000000E+00		
7		.130000000000E+02	.119781250000E+03	.354371903869E-08	.257903736774E+01		
8		.621937215328E-05	.135628201533E-01	.103395432234E-04	.515377099800E+04		
9		.122384000000E+06	.139698386192E-06	.312475661845E+01	.149011611938E-07		
10		.980059657704E+00	.185781250000E+03	-.149515449880E+01	-.732566228610E-08		
11		.260725145963E-10	.100000000000E+01	.219900000000E+04	.000000000000E+00		
12		.240000000000E+01	.000000000000E+00	-.111758708954E-07	.130000000000E+02		
13		.121596000000E+06	.400000000000E+01				
14	28 22	2 28 10 0 0.0	.390788540244E-03	-.818545231596E-11	.000000000000E+00		
15		.900000000000E+01	-.534375000000E+01	.419196032626E-08	.307835828733E+01		
16		-.258907675743E-06	.167768010870E-01	.899843871593E-05	.515369073486E+04		
17		.122400000000E+06	.206753611565E-06	.210962065149E+01	.987201929092E-07		
18		.969036079899E+00	.208437500000E+03	-.128598749440E+01	-.746423948696E-08		
19		-.715744099329E-09	.100000000000E+01	.219900000000E+04	.000000000000E+00		
20		.240000000000E+01	.630000000000E+02	-.111758708954E-07	.900000000000E+01		
21		.121596000000E+06	.400000000000E+01				
22	14 22	2 28 10 0 0.0	-.889501534402E-04	-.409272615798E-11	.000000000000E+00		
23		.620000000000E+02	-.121875000000E+02	.468519515718E-08	-.840438501671E+00		
24		-.711530447006E-06	.147903268225E-02	.866875052452E-05	.515364384460E+04		
25		.122400000000E+06	.502914190292E-07	.205603914244E+01	-.104308128357E-06		
26		.953321908358E+00	.212281250000E+03	.302557863879E+01	-.797783230874E-08		
27		-.704672209569E-09	.100000000000E+01	.219900000000E+04	.000000000000E+00		
28		.240000000000E+01	.000000000000E+00	-.745058059692E-08	.318000000000E+03		
29		.121596000000E+06	.400000000000E+01				

Figure 3.8 RINEX nav file

Data Observation file

RINEX Obs file [o] provides functionality to read and write Global Navigation Satellite System (GNSS) observation data. The file contains pseudorange data in ASCII format that

conforms to the RINEX standard. It is GPS Time, and the scalar data is not referenced by any particular frame. In obs file date, timing of data collection with carrier and code pseudoranges with organized corresponding to the related satellites PRN number and other information of every satellite in view is given [52]. Further details can be seen in figure 3.9.

1	2.10	OBSERVATION DATA	G (GPS)	RINEX VERSION / TYPE
2	RTKCONV 2.4.2		20220301 060919 UTC	PGM / RUN BY / DATE
3	log: E:\check\easy1-6\easy (1-6)\Field Data\RINEX\2nd rove			COMMENT
4	format: RINEX			COMMENT
5				MARKER NAME
6				MARKER NUMBER
7				OBSERVER / AGENCY
8				REC # / TYPE / VERS
9				ANT # / TYPE
10	1540958.3492	5095210.4558	3502614.5733	APPROX POSITION XYZ
11	0.0000	0.0000	0.0000	ANTENNA: DELTA H/E/N
12	1	1		WAVELENGTH FACT L1/2
13	10	C1	L1	D1 S1 P1 C2 L2 D2 S2# / TYPES OF OBSERV
14		E2		# / TYPES OF OBSERV
15	2022	2	28	9 46 11.0040000 GPS TIME OF FIRST OBS
16	2022	2	28	9 56 18.0050000 GPS TIME OF LAST OBS
17				END OF HEADER
18	22	2	28	9 46 11.0040000 0 9G13G15G17G28G30G14G19G 1G24
19	24059681.864	126434479.129	-2220.150	42.000
20				
21	25604455.872	134552327.7111	-1436.511	44.000
22				
23	21780778.871	114458766.9961	-1040.811	40.000
24				

Figure 3.9 RINEX obs file

To implement the RAIM algorithm different MATLAB functions are used and some modifications are made in the coding to get the required results. Following functions are used: rinex function, respo_ls, grabdata, fephoch_2 etc.

Input to the RAIM Fault Detection and Exclusion Algorithm

The required input for the algorithm are:

- ❖ Geometry matrix
- ❖ Error matrix
- ❖ Pseudorange observation

- ❖ Variance
- ❖ Probabilities of false alarm $P_{(FA)}$ and missed detection $P_{(MD)}$

The output of the RAIM Fault Detection and Exclusion Algorithm

The output taken from the RAIM algorithm are:

- ❖ HPL
- ❖ Faulty measurements
- ❖ Position after exclusion

4. Results and Discussion

4.1 Experiment and Results of Dataset-1

Dataset-1 was collected in the IST Khatana Stadium on 28th February 2021 under a clear sky and the total number of available satellites was 9. For algorithm testing, we intentionally introduced an error of 60m in the observation measurements of PRN 17 at epoch 400 and onward. When error was added in one observation it tends to increase several values but the largest value of the measurement that is higher than the threshold is considered an outlier. For detection and identification, thresholds are calculated using the *F*-distribution and normal distribution using the probability of false alarm and missed detection. For which significance level α and α_o are predefined 0.1 % and value of β is chosen 80% as in [53]

When the RAIM algorithm is applied, the measurements are linearized using equation (4) to calculate measurement residual matrix which is used to calculate the test statistics for the fault detection and identification of outlier. For fault detection global test is applied and when the test statistics exceeded the threshold an alarm is generated. The results of dataset-1 for fault detection are clearly seen in figure 4.1 at epoch 400 suddenly test statistic value of PRN 17 increases.

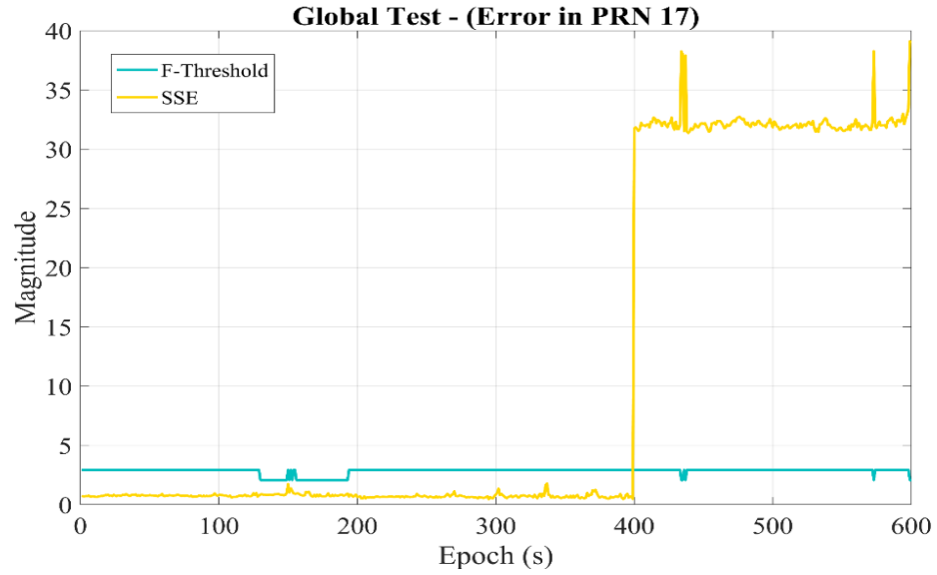


Figure 4.1 Global Test (Dataset -1)

For detection of erroneous measurement, protection level can also be set as a threshold and compared with test statistics Sum of Squared Residuals (SSE) and alarm is generated when SSE exceeds the protection level as shown in figure 4.2.

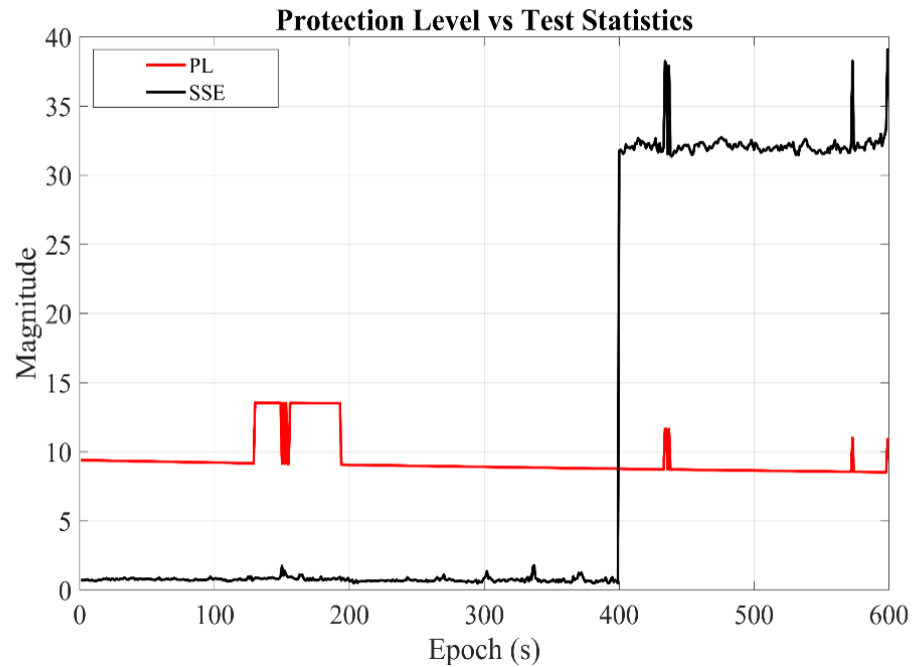


Figure 4.2 Protection level vs Test Statistics

Once the erroneous measurements were detected the next step is to identify those measurements. For this local test based on LSR testing and correlation analysis method was applied in which the test statistics for the local test were found to be exceeding the threshold at the epoch in which error was introduced as seen in Figure 4.3. The residuals and global test values for epochs 399 and 400 are given in table 4.1 and values of PRN 17 are the largest values.

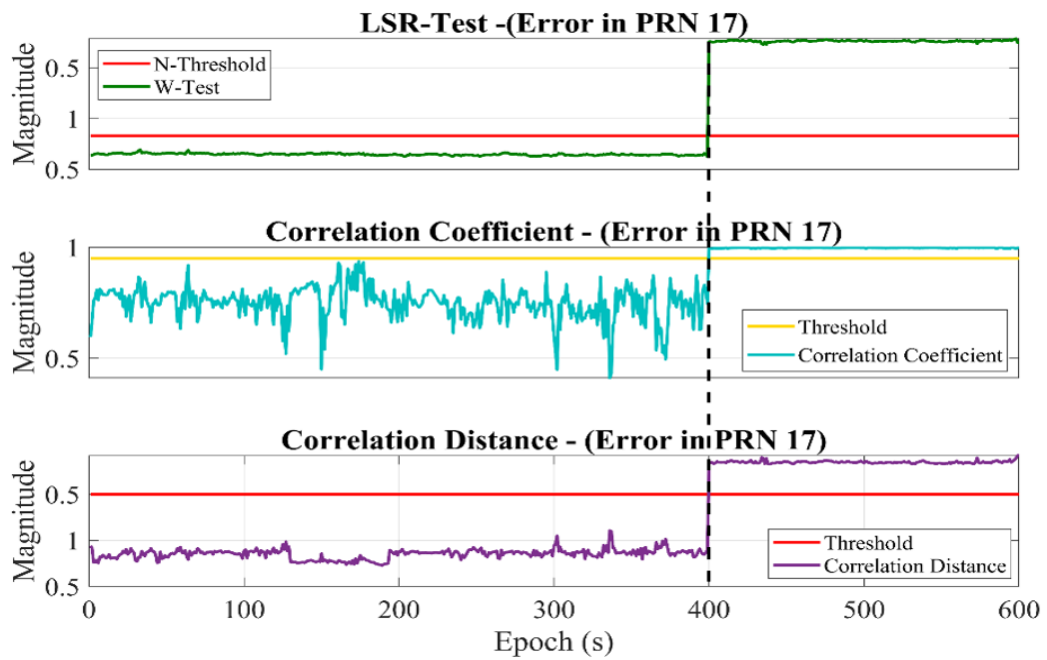


Figure 4.3 Fault Identification test: LSR- Test, Correlation Coefficient, and Correlation Distance (Dataset-1)

When the results of all the three identification tests are same, the erroneous measurement is excluded from the measurements while calculating the position. For before and after exclusion of faulty measurement, the results can be compared in Figure 4.4. In Figure 4.4 in left subplot, the gap shows the presence of an outlier while after applying RAIM algorithm the faulty measurement is removed and we get a reliable position.

Table 4.1 Residual and Fault identification values for error introduced at epoch 400 for dataset-1

PRN	Epochs	Residuals	LSR Test	Correlation Coefficient	Distance Correlations
PRN 13	399	-0.3626	0.1279	0.1728	5.2737
	400	-2.0241	0.7139	0.0803	19.65
PRN 15	399	3.5785	0.7997	0.5477	6.969
	400	6.1679	1.3782	0.1028	19.70
PRN 17	399	4.5959	1.4416	0.7459	2.800
	400	46.219	14.497	0.9961	14.74
PRN 28	399	-0.6612	0.2041	0.0786	5.082
	400	-16.8802	5.2112	0.3561	20.72
PRN 30	399	0.4974	0.3031	0.0949	3.360
	400	5.0501	3.0771	0.2004	18.51
PRN 14	399	0.3301	0.1010	0.1006	4.769
	400	-14.126	4.3199	0.2809	20.48
PRN 19	399	-4.3304	1.4070	0.7457	6.153
	400	-22.44	7.2941	0.4942	21.13
PRN 1	399	-1.5076	0.5821	0.2482	5.147
	400	0.1920	0.0741	0.0149	19.31
PRN 24	399	-1.4017	0.3065	0.2959	9.481
	400	2.4872	0.5440	0.0210	20.47

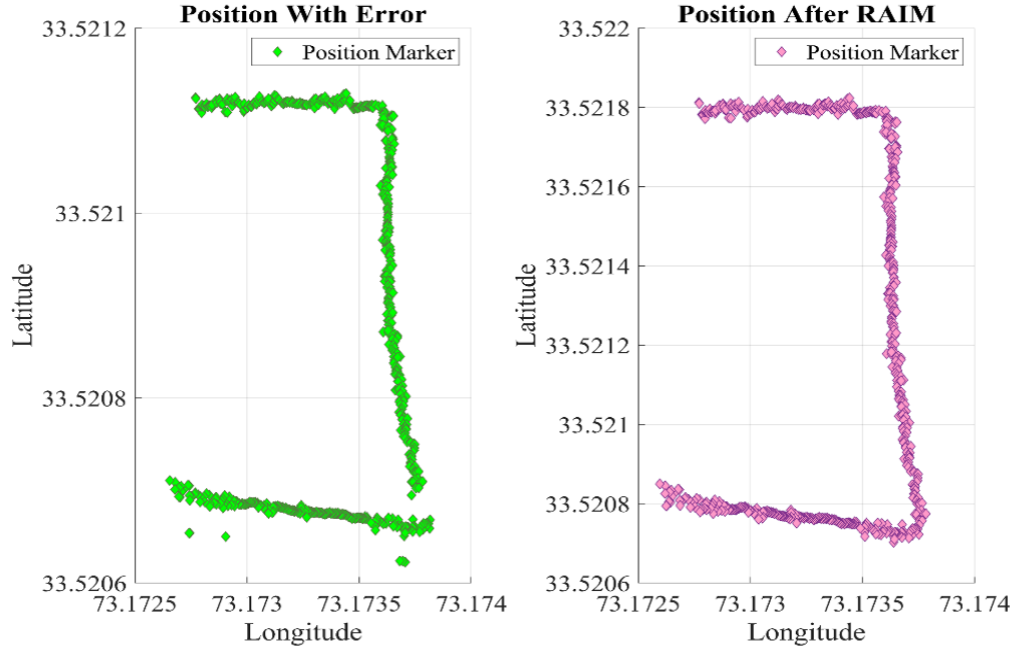


Figure 4.4 Position with and without error

4.2 Experiment and Results of Dataset -2

In dataset 2 the number of available satellites is 11 and the number of epochs is 473. Error is intentionally introduced at epoch 150 and onward. Same as for dataset 1 RAIM algorithm was applied for dataset 2. For fault detection global test is applied and when the test statistics exceeded the threshold an alarm is generated. This can be seen in figure 4.5 where the test statistics are compared with the threshold. Before the epoch 150, the test statistics lie below the specified threshold while at epoch 150 (where the error was added) the test statistics value suddenly jumps beyond the threshold. This is the indication of an erroneous epoch.

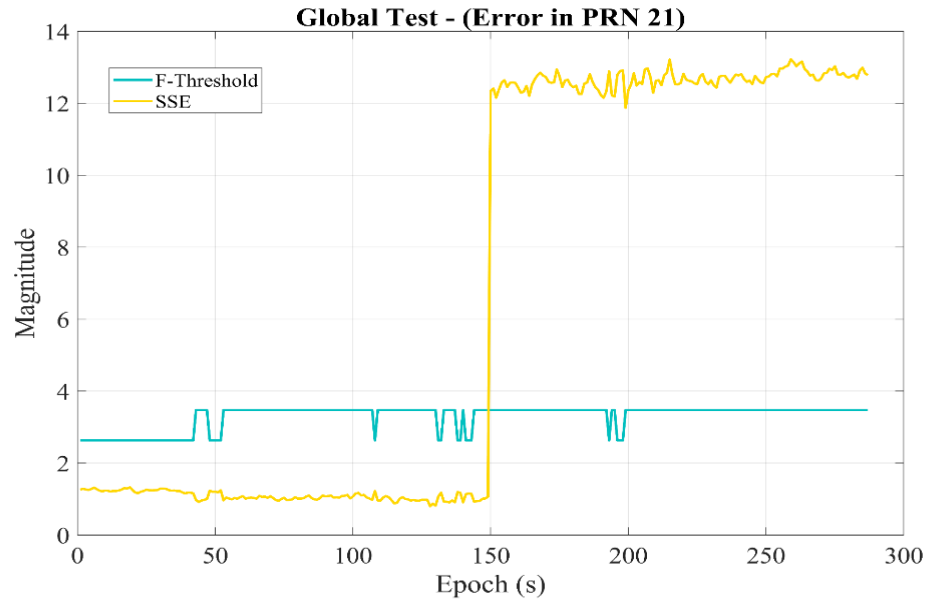


Figure 4.5 Global Test (Dataset-2)

Same as for dataset 1 protection level can also be used as a threshold for the detection of erroneous measurement and compared with test statistics SSE, an alarm is generated when SSE exceeds the protection level at epoch 150 and onwards as shown in figure 4.6.

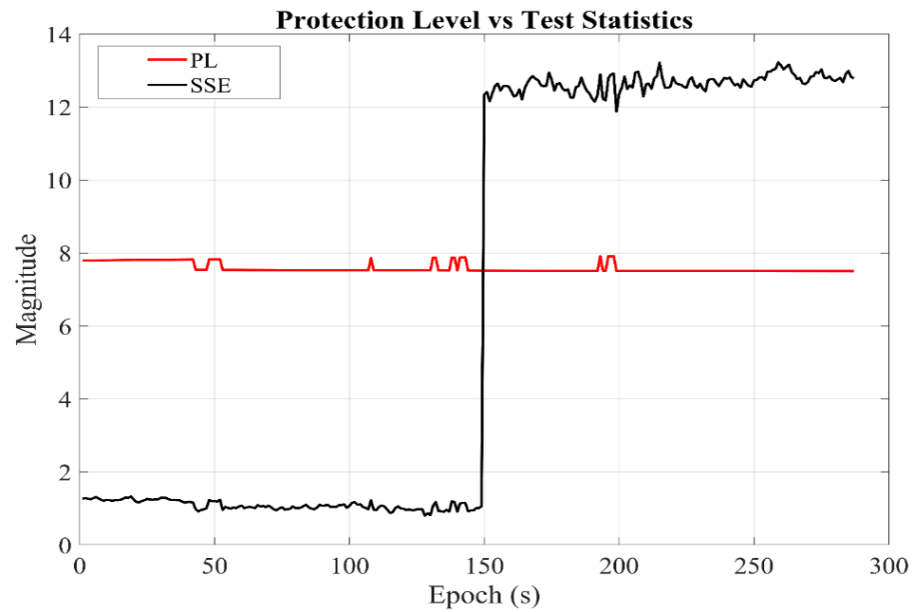


Figure 4.6 Protection Level vs Test Statistics (Dataset-2)

After detection, the next step is to identify those measurements. For this, local test based on LSR testing and correlation analysis method was applied in which the test statistics for local test were found to be exceeding the threshold at the epoch in which error was introduced as seen in figure 4.7. In this figure, three different tests are applied at the same time which results in the same erroneous measurement.

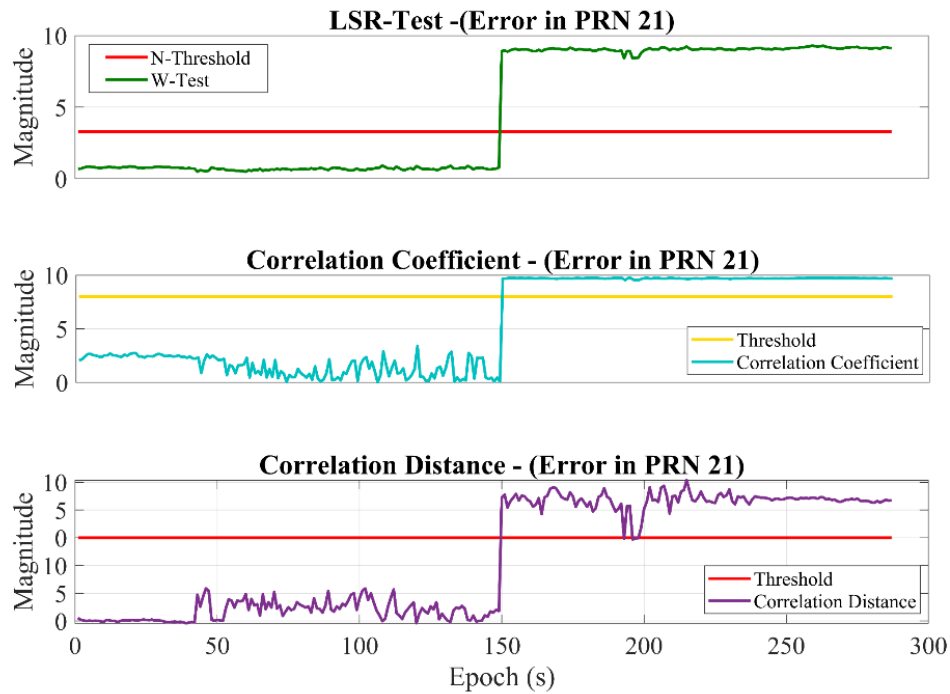


Figure 4.7 Fault Identification test: LSR- Test, Correlation Coefficient and Correlation Distance (Dataset-2)

Last step after identification was the exclusion of erroneous measurement from calculation while calculating the position. For before and after exclusion of faulty measurement, the results can be compared in figure 4.8 and difference is clearly seen. On left hand side, the missing points in the track represents the epoch where the error was added. However, after excluding the erroneous measurement, the error is neutralized as seen on the right side of the figure.

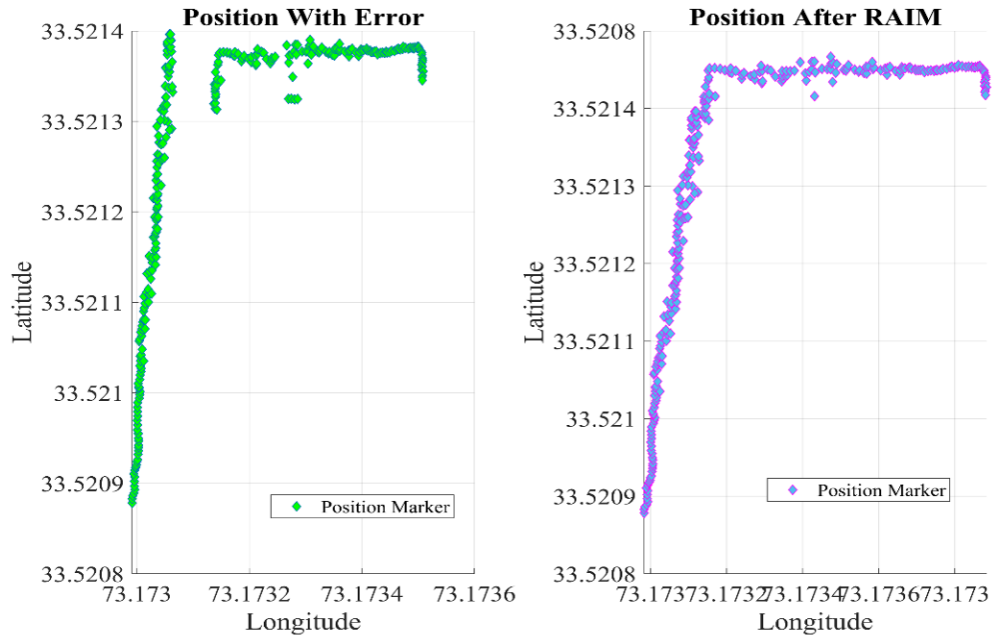


Figure 4.8 Position with and without Error

4.3 Experiment and Results of Dataset-3

The above two datasets were dynamic. For further verification of the working of the algorithm, we collected static datasets and applied the RAIM algorithm. The goal was to see the behavior of the algorithm in different conditions and different modes (i.e. static or dynamic). After the application of the algorithm to the static dataset, the results were found to be same as for the dynamic situations. During this dataset, a total number of 12 GPS L1 C/A satellites were tracked and the data was recorded for 473 epochs. Figure 4.9, 4.10, and 4.11 represents the results for the global test, protection levels, and identification tests respectively. While the position calculation can be seen in figure 4.12.

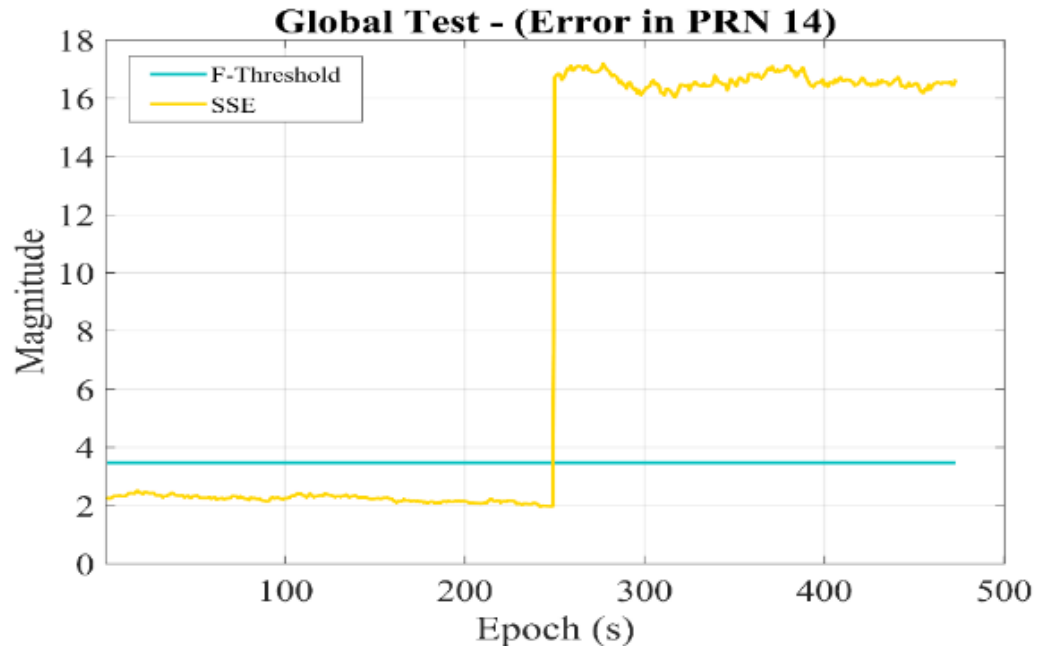


Figure 4.9 Global Test (Dataset-3)

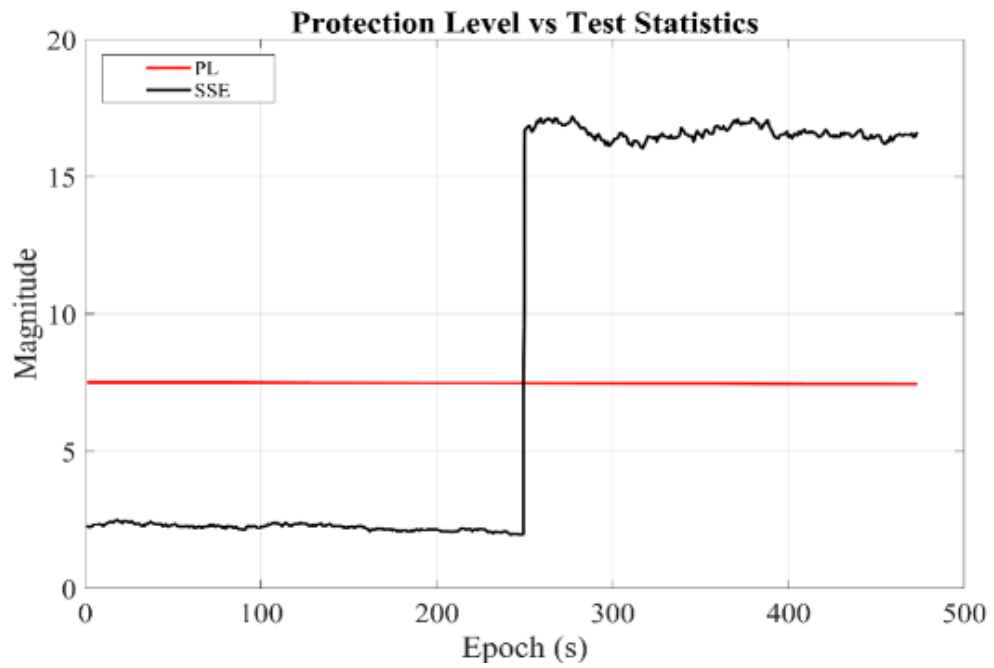


Figure 4.10 PL vs SSE (Dataset-3)

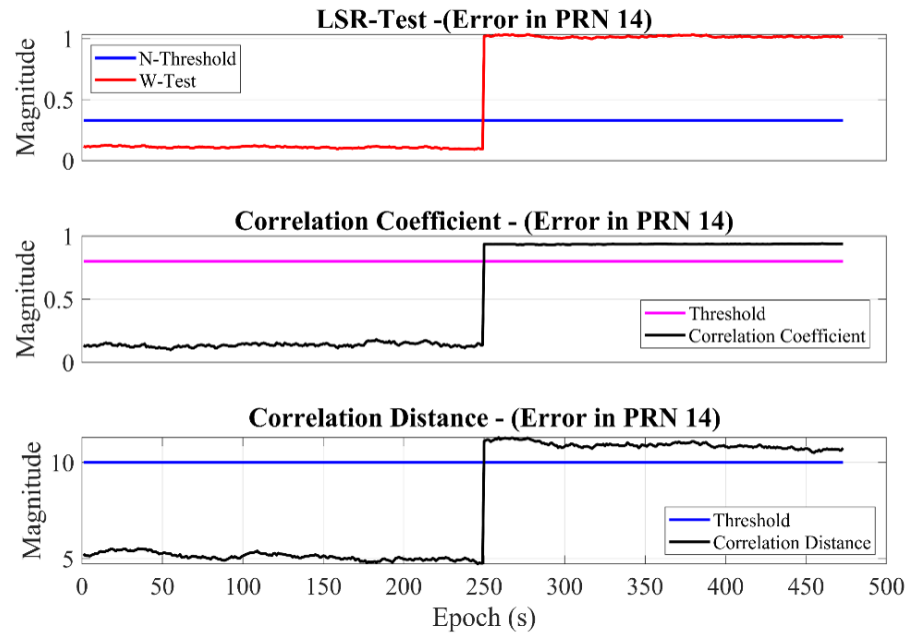


Figure 4.11 Fault Identification Tests (Dataset-3)

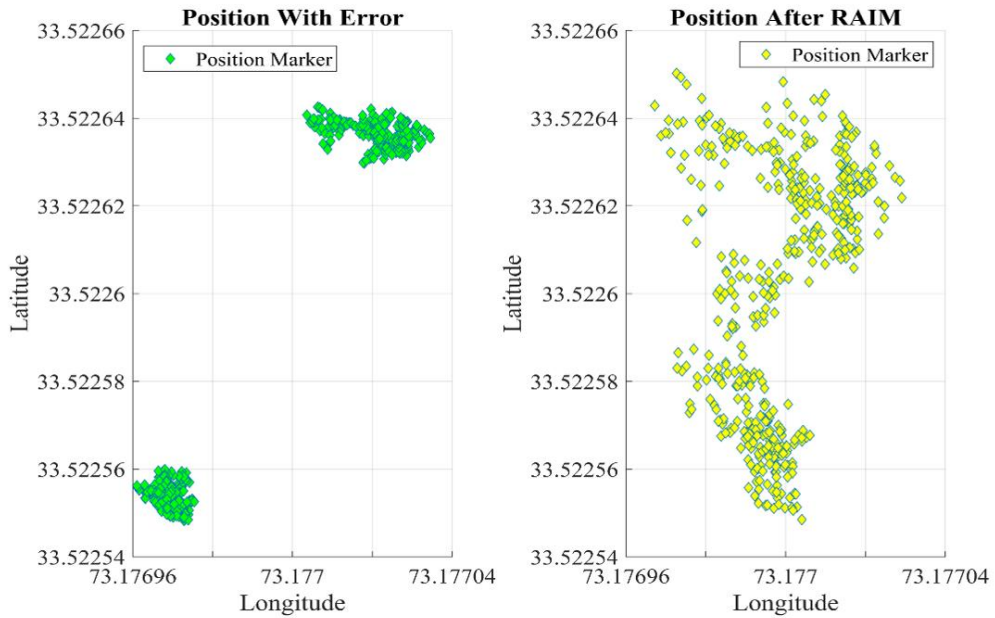


Figure 4.12 Position with error and after applying RAIM

During the algorithm results analysis, an error of different magnitude was intentionally introduced, and we analyze that when the error magnitude was greater than 20m the

detection and identification rate was 100% however, when the error magnitude was less than 20m the detection and identification rate of LSR test was decreasing and missed detection and false alarm was generated. Figure 4.13 & 4.14 shows the detection and identification results of datasets 1 and 2 respectively. It can be seen that LSR test failed to identify the erroneous measurement when error magnitude less than 20m at significance level α and α_0 are 0.1 % while correlation coefficient and correlation distance identifying the erroneous measurement.

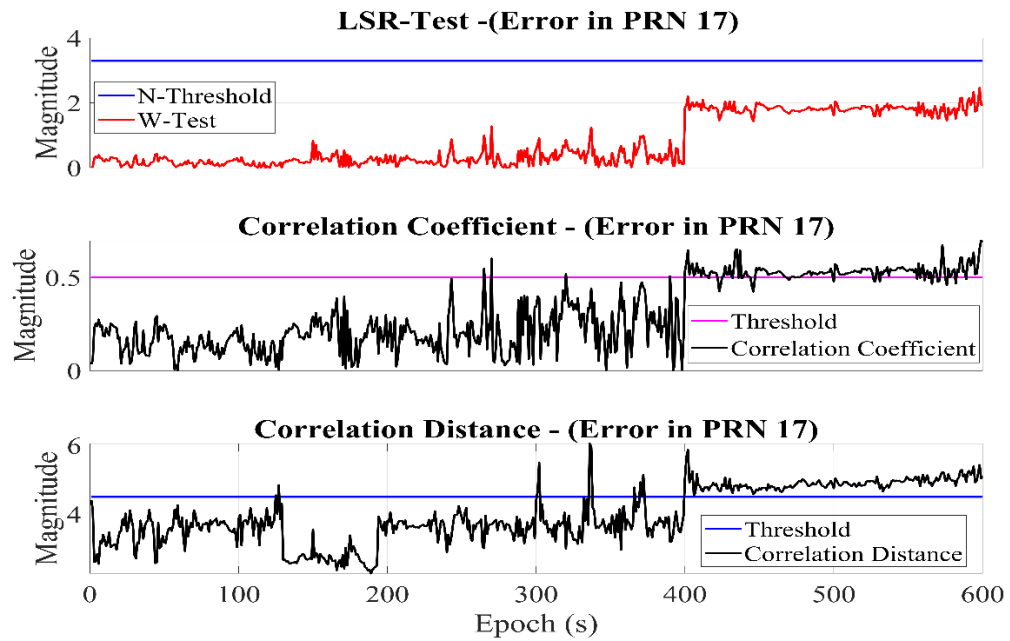


Figure 4.13 Results of identification tests when error magnitude is less than 20m.(Dataset-1)

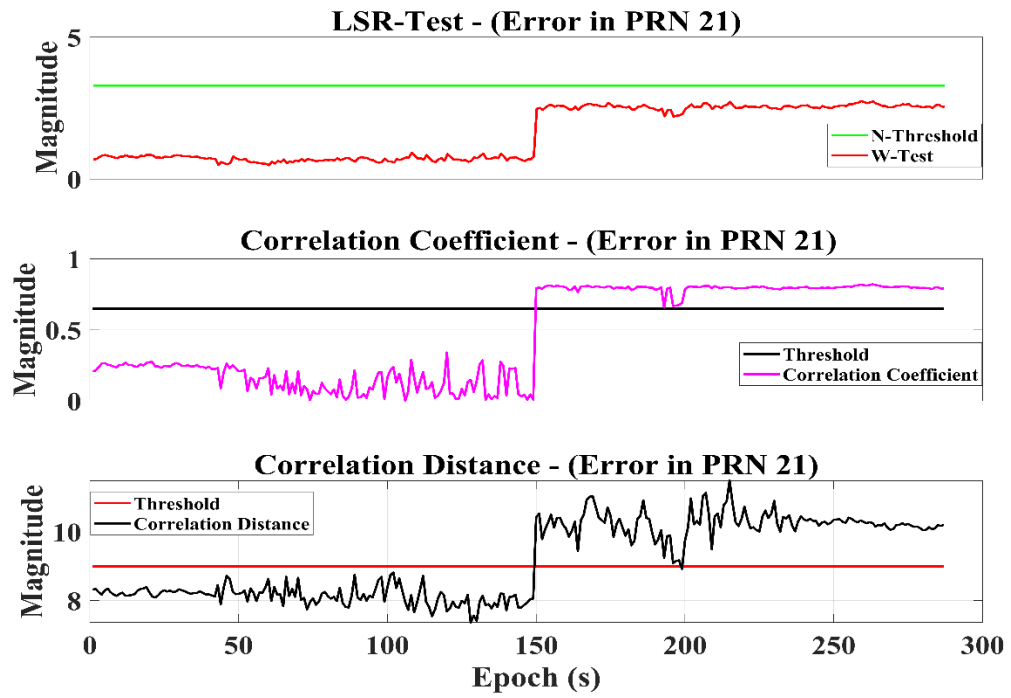


Figure 4.14 Results of identification tests when error magnitude is less than 20m.(Dataset-2)

5. Summary and Future Research Direction

5.1 Summary

Single frequency GNSS receivers are predominantly available in market and are cost effective, making them suitable for commercial applications. However, the position and navigation solution provided by these receivers can suffer severely even if one faulty measurement is present in the position calculation. Therefore, such satellite needs to be identified and excluded before calculating the user position to ensure better accuracy and reliable output. For this purpose, we implemented a receiver autonomous integrity monitoring algorithm using least square residual and correlation analysis method for identification of faulty measurements. The global test is used for the detection of faulty measurement and a polling scheme based on three local tests is used for its identification.

The local test depends upon the redundancy due to which the threshold varies as a result of change in the number of satellites. To ensure the correct identification by local test, correlation analysis method is used. The correlation method further consists of two steps i.e., correlation coefficient calculation and correlation distance. The threshold of the correlation distance varies with the dataset. Therefore, we are using three tests to compare the results and identify the faulty measurement based on these results. The global tests raise an alarm whenever it detects faulty measurements. After that the local test, backed by the correlation analysis identifies the faulty measurement which is then excluded from the position calculation algorithm. Results of both before and after the exclusion of the faulty measurement can be seen in the figures which confirms the working of this algorithm. The algorithm mentioned in this paper does not only generates a timely warning upon the detection of erroneous measurement, but it also removes the satellite from the

measurements which ensures a reliable output. Moreover, the algorithm introduced in this paper is straightforward and easy to implement. Thus, it can be very helpful for improving the performance of low-cost single frequency GNSS receivers.

5.2 Future Research Direction

- ❑ Further research work is needed to enhance this algorithm for multiple satellite failure.
- ❑ This algorithm can be combined with fused sensor-based navigation to improve the fault exclusion principle. Intelligent decision-making algorithm can be added to make the navigating body decide when to switch from GNSS to alternate positioning method (in case the number of satellites falls short).
- ❑ RAIM can also be implemented for the detection of GNSS spoofing.

6. References

- [1] E. D. Kaplan and C. J. Hegarty, *Understanding GPS/GNSS Principles and Applications*. Artech house, 2017. [Online]. Available: <http://ieeexplore.ieee.org/document/9100468>
- [2] P. D. Groves, *Principles of GNSS , Inertial , and Multisensor Integrated*. 2008.
- [3] B. Hofmann-Wellenhof, H. Lichtenegger, and E. Wasle, *GNSS-Global Navigation Satellite Systems: GPS, GLONASS, Galileo, and more*, vol. 45, no. 11. SpringerWienNewYork, 2008. doi: 10.5860/choice.45-6185.
- [4] P. Misra and P. Enge, “Global Positioning System: Signals, Measurements, and Performance - Revised Second Edition (2011),” *International Journal of Wireless Information Networks*. Ganga Jamuna Press, p. 115, 2011. [Online]. Available: <https://books.google.com/books?id=pv5MAQAAIAAJ>
- [5] Teunissen, Peter JG, and Oliver Montenbruck, eds. *Springer handbook of global navigation satellite systems*. Vol. 10. New York, NY, USA: Springer International Publishing, 2017. doi: 10.1007/978-3-319-42928-1.
- [6] U. Force, “NAVSTAR Global Positioning System Interface Specification (IS-GPS-200 Rev. D),” *Interface Specif. IS-GPS-200D, US Air Force*, no. December, 2006.
- [7] B. W. P. and S. W. Gilbert., “NAVSTAR: Global positioning system – Ten years later.” *Proceedings of the IEEE* 71(10), pp. 1177–1186., 1983.
- [8] V. V. D. Y. I. N. Y. M. U. A. M. Finkel’shtein., “The Russian global navigation satellite program,” *Her. Russ. Acad. Sci*, 2009, pp. 79(1), 7–13.
- [9] D. Blonski., “Performance Extrapolation to FOC and outlook to Galileo early services,” 2014.
- [10] C. H. Y. Y. Z. Cai., “BeiDou Navigation Satellite System and its time scales,” *Metrologia* 48(4), 2011, pp. S213–S218.
- [11] M. S. J. T. T. Okamoto., *Establishment of regional navigation satellite system utilizing quasi-zenith satellite system*. Mitsubishi Electr. Adv. Mag, 2014.

- [12] H. H. M. R. S. D. Badnore., “Indian Regional Navigation System,” *Int. J. Res. Sci. Eng.* 1(SP1), 2015, pp. 36–42.
- [13] E. Cohen, “GNSS Position Integrity in Urban Environments: A Review of Literature,” *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 9, pp. 2762–2778, 2018, doi: 10.1109/TITS.2017.2766768.
- [14] A. Angrisano, S. Gaglione, and C. Gioia, “RAIM algorithms for aided GNSS in Urban scenario,” *Ubiquitous Positioning, Indoor Navig. Locat. Based Serv. UPINLBS*, 2012, doi: 10.1109/UPINLBS.2012.6409786.
- [15] “GNSS Positioning Techniques | Tallysman,” 2021. <https://www.tallysman.com/gnss-positioning-techniques/> (accessed Jun. 24, 2022).
- [16] K. Borre, D. M. Akos, N. Bertelsen, P. Rinder, and S. H. Jensen, “A software-defined GPS and galileo receiver: A single-frequency approach,” *Appl. Numer. Harmon. Anal.*, no. 9780817643904, pp. 1–168, 2007.
- [17] M. Karaim, M. Elsheikh, and A. Noureldin, “GNSS Error Sources,” *Multifunct. Oper. Appl. GPS*, 2018, doi: 10.5772/intechopen.75493.
- [18] H. Wang, Y. Cheng, C. Cheng, S. Li, and Z. Li, “Research on satellite selection strategy for receiver autonomous integrity monitoring applications,” *Remote Sens.*, vol. 13, no. 9, 2021, doi: 10.3390/rs13091725.
- [19] M. Tahsin, S. Sultana, T. Reza, and M. Hossam-E-Haider, “Analysis of DOP and its preciseness in GNSS position estimation,” Oct. 2015. doi: 10.1109/ICEEICT.2015.7307445.
- [20] EUSPA, *EUSPA EO and GNSS Market Report*, no. 1. 2022. doi: 10.2878/94903.
- [21] “GNSS Performances - Navipedia,” *Navipedia*, 2014. https://gssc.esa.int/navipedia/index.php/GNSS_Performances (accessed Jun. 24, 2022).
- [22] Q. Sun and J. Zhang, “RAIM method for improvement on GNSS reliability and integrity,” *AIAA/IEEE Digit. Avion. Syst. Conf. - Proc.*, pp. 1–11, 2009, doi:

10.1109/DASC.2009.5347420.

- [23] J. Wang and P. B. Ober, “On the availability of fault detection and exclusion in GNSS receiver autonomous integrity monitoring,” *J. Navig.*, vol. 62, no. 2, pp. 251–261, 2009, doi: 10.1017/S0373463308005158.
- [24] B. W. Parkinson and S. Telecom, *Global Positioning System: Theory and Applications, Volume I*, vol. 2. 1996. doi: 10.2514/4.866388.
- [25] A. Grosch, O. G. Crespillo, I. Martini, and C. Günther, “Snapshot residual and Kalman Filter based fault detection and exclusion schemes for robust railway navigation,” *2017 Eur. Navig. Conf. ENC 2017*, pp. 36–47, 2017, doi: 10.1109/EURONAV.2017.7954171.
- [26] H. Kuusniemi, “User-Level Reliability and Quality Monitoring in Satellite-Based Personal Navigation,” *Technology*, no. June, p. 220, 2005.
- [27] R. G. BROWN, “A Baseline GPS RAIM Scheme and a Note on the Equivalence of Three RAIM Methods,” *Navigation*, vol. 39, no. 3, pp. 301–316, 1992, doi: 10.1002/j.2161-4296.1992.tb02278.x.
- [28] D. M. Hawkins, *Identification of Outliers*. 1980. doi: 10.1007/978-94-015-3994-4.
- [29] Y. Lee, “Analysis of range and position comparison methods as a means to provide GPS integrity in the user receiver,” *Navig. J. Inst. Navig.*, pp. 1–17, 1986, [Online].
- [30] B. W. PARKINSON and P. AXELRAD, “Autonomous GPS Integrity Monitoring Using the Pseudorange Residual,” *Navigation*, vol. 35, no. 2, pp. 255–274, 1988, doi: 10.1002/j.2161-4296.1988.tb00955.x.
- [31] M. A. STURZA, “Navigation System Integrity Monitoring Using Redundant Measurements,” *Navigation*, vol. 35, no. 4, pp. 483–501, 1988, doi: 10.1002/j.2161-4296.1988.tb00975.x.
- [32] Y. C. Lee, “Receiver Autonomous Integrity Monitoring (RAIM) Capability for Sole-Means GPS Navigation in the Oceanic Phase of Flight,” *IEEE Aerosp. Electron. Syst. Mag.*, vol. 7, no. 5, pp. 29–36, 1992, doi: 10.1109/62.257089.

- [33] S. Hewitson and J. Wang, “GNSS receiver autonomous integrity monitoring (RAIM) performance analysis,” *GPS Solut.*, vol. 10, no. 3, pp. 155–170, Jul. 2006, doi: 10.1007/s10291-005-0016-2.
- [34] A. Rakipi, B. Kamo, S. Cakaj, V. Kolici, A. Lala, and I. Shinko, “Integrity Monitoring in Navigation Systems: Fault Detection and Exclusion RAIM Algorithm Implementation,” *J. Comput. Commun.*, vol. 03, no. 06, pp. 25–33, 2015, doi: 10.4236/jcc.2015.36004.
- [35] J. Blanch, T. Walter, P. Enge, and V. Kropp, “A simple position estimator that improves advanced RAIM performance,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 3, pp. 2485–2489, 2015, doi: 10.1109/TAES.2015.140674.
- [36] M. Joerger and B. Pervan, “Kalman filter-based integrity monitoring against sensor faults,” in *Journal of Guidance, Control, and Dynamics*, 2013, vol. 36, no. 2, pp. 349–361. doi: 10.2514/1.59480.
- [37] J. Blanch *et al.*, “Baseline advanced RAIM user algorithm and possible improvements,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 1, pp. 713–732, 2015, doi: 10.1109/TAES.2014.130739.
- [38] X. Ma, K. Yu, J. P. Montillet, and X. He, “Equivalence Proof and Performance Analysis of Weighted Least Squares Residual Method and Weighted Parity Vector Method in RAIM,” *IEEE Access*, vol. 7, pp. 97803–97814, 2019, doi: 10.1109/ACCESS.2019.2929073.
- [39] L. Jiang and Z. Xiaolin, “GNSS Fault Detection and Exclusion Based on Virtual Pseudorange-Based Consistency Check Method,” *Chinese J. Electron.*, vol. 29, no. 1, pp. 41–48, Jan. 2020, doi: 10.1049/CJE.2019.09.005.
- [40] T. Yang, D. S.-I. T. on A. and, and undefined 2019, “Global navigation satellite systems fault detection and exclusion: A parameterized quadratic programming approach,” *ieeexplore.ieee.org*, Accessed: Jun. 24, 2022. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/8917608/>
- [41] H. Imtiaz, M. Haroon Yousaf, H. Sajjad Malik, and M. Ali Akhtar, “Design and

- implementation of receiver autonomous integrity monitoring algorithm on DSP for small UAV applications,” in *Proceedings - 2019 International Conference on Frontiers of Information Technology, FIT 2019*, 2019, pp. 278–283. doi: 10.1109/FIT47737.2019.00059.
- [42] S. Z. Farooq, D. Yang, and E. N. J. Ada, “A cycle slip detection framework for reliable single frequency RTK positioning,” *Sensors (Switzerland)*, vol. 20, no. 1, 2020, doi: 10.3390/s20010304.
- [43] J. Xu *et al.*, “Design and implementation of channel estimation and equalization of indoor positioning system based on UWB,” *2009 2nd Int. Conf. Intell. Comput. Technol. Autom. ICICTA 2009*, vol. 4, pp. 57–61, 2009, doi: 10.1109/ICICTA.2009.730.
- [44] R. Sabatini, T. Moore, and S. Ramasamy, “Global navigation satellite systems performance analysis and augmentation strategies in aviation,” *Progress in Aerospace Sciences*, vol. 95, no. August. Elsevier Ltd, pp. 45–98, 2017. doi: 10.1016/j.paerosci.2017.10.002.
- [45] E. Wang, F. Yang, and T. Pang, “GNSS receiver autonomous integrity monitoring algorithm based on least squared method,” in *Proceedings of the 2017 12th IEEE Conference on Industrial Electronics and Applications, ICIEA 2017*, 2018, vol. 2018-Febru, no. 1, pp. 2017–2021. doi: 10.1109/ICIEA.2017.8283169.
- [46] R. G. Brown and G. Y. Chin, “GPS RAIM: Calculation of threshold and protection radius using chi-square methods,” *Glob. Position. Syst. (Red B. Ser.)*, vol. 4.
- [47] X. Zheng, Y. Liu, G. Fan, J. Zhao, and C. Xu, “Analyses of the sensitivity of multi-constellation advanced receiver autonomous integrity monitoring vertical protection level availability to error parameters and a failure model over China,” *Adv. Mech. Eng.*, vol. 10, no. 6, pp. 1–11, 2018, doi: 10.1177/1687814018776191.
- [48] J. Zhao, D. Song, C. Xu, and X. Zheng, “A Modified LSR Algorithm Based on the Critical Value of Characteristic Slopes for RAIM,” *IEEE Access*, vol. 7, pp. 70102–70116, 2019, doi: 10.1109/ACCESS.2019.2917377.

- [49] Q. Zhang, L. Zhao, and J. Zhou, "Improved Method for Single and Multiple GNSS Faults Exclusion based on Consensus Voting," *J. Navig.*, vol. 72, no. 4, pp. 987–1006, 2019, doi: 10.1017/S0373463318001133.
- [50] B. Hu, Y.F., Lu, J.J., Cao, K.J. and Li, "RAIM algorithm based on correlation distance with several epoch.," *J. Detect. Control*, vol. 36, pp. 42–45, 2014.
- [51] B. Jinzhon, G. Shouzhou, and F. Shushan, "Studied on a New RAIM Method Based on Vector Correlation Distances," *Sci. Sin. (Physica, Mech. \& Astron.*, vol. 40, pp. 1–7, 2010.
- [52] W. Zhang, "GPS Short-Distance Baseline Estimation from RINEX Files under Matlab Environment," 2005.
- [53] W. Baarda, "A Testing Procedure for Use in Geodetic Networks," *Publ. Geod.*, vol. 2, no. 5, p. 97, 1968.